جامعة الملك سعود
King Saud University
كلية علوم الحاسب والمعلومات
قسم تقنية المعلومات

لسانٌ فصيحٌ إذا صَحَّ النُّطقُ فيه

IT 496: Graduation Project Report
Sprint-1
فصـــيح

Prepared by

Name	ID
Tala Alsheail	444200518
Walah Alsaeed	444201689
Norah Alkathiri	444200439
Joud Alzahrani	444200862

Supervised by
Dr. Reem Alqifari

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1 Introduction

The Arabic language is important in many aspects; it is one of the most widely spoken languages, with approximately 380 million speakers worldwide [1]. It also serves as the language of Islam, and it was ranked to be the 4th most useful language for business [2][3]. Despite its significance, Arabic is often perceived as a hard language to learn due to factors such as script complexity, pronunciation, and unique letters. These factors can make the learning process challenging for many learners.

The problem we address in this project is the lack of platforms that support the improvement of Arabic pronunciation in an interactive and personalized manner. Current teaching methods (e.g. School Textbooks, Direct Translation) usually offer a more general approach, focusing on overall content delivery (Grammar, Calligraphy, Reading, etc) rather than targeting a specific area. This causes an absence of feedback, interactivity, pronunciation support and mispronunciation identification [4]. The lack of these factors can cause the learning process to be more challenging because individual learner needs are not adequately addressed and opportunities to leverage modern technological approaches remain underutilized.

Studies have shown that integrating a technological approach into Arabic language education is a powerful opportunity to improve the learner's skill [5]. Locally, it supports the currently existing traditional teaching methods, which can lack user engagement and interactivity, by introducing an AI-driven approach. This enables adaptive learning tailored to different users and aligns with the Saudi vision 2030 goal of upholding the Arabic language [6], aiming to strengthen the Islamic and national identity. Globally, it will serve as a new gateway for individuals to learn Arabic and further support the spread of the language. It also contributes to the development of Arabic language learning technologies, which remain limited compared to other languages, while supporting AI advancement for low-resource languages such as Arabic. In addition, the system can be extended to non-native Arabic learners worldwide, encouraging personalized AI-based learning approaches in education, and provides a model for pronunciation learning systems that can be adapted to other languages.

This document represents the comprehensive report of our proposed AI-driven system, supporting the Arabic pronunciation improvement process for Arabic-speaking children, by offering interactive exercises to improve their pronunciation, according to the results of an assessment test which was designed to specifically identify their struggle points with pronunciation. We will present the identified problem, proposed solution, product vision, product roadmap, objectives to achieve, scope, hardware/software tools, and team responsibilities. Additionally, we provide the necessary background on Arabic pronunciation learning, phonetic challenges, speech processing, and automated pronunciation assessment, followed by a literature review of related work, dataset exploration, and competitive product analysis.

Lastly, the document explains the system requirements, including user analysis, requirement elicitation, user interactions, and the product backlog. It also covers different system design aspects through various representative diagrams. The implementation section describes the major development steps, system components, API and third-party service integration, core functionalities, AI-based speech analysis, audio preprocessing, and model-building experiments. Finally, the report presents the testing process, including experimental results and user acceptance testing, and concludes with the project's impact, challenges, limitations, contributions, and future work.

1.1 The Problem

Arabic pronunciation presents a significant real-world challenge for individual learners, the language includes unique phonetic characteristics such as deep throat sounds and emphatic letters that are difficult to articulate correctly without continuous guidance[7]. A lot of learners face difficulties to differentiate and pronounce letters such as ر،ق،ج،ك،خ،غ،ض correctly which often leads to unclear speech or unintended changes in meaning[8]. These difficulties are further compounded by incorrect distinction between short and long vowels, and inaccurate stress placement within words, which often result in misreading and incorrect pronunciation.

For example, the word **"قلب"** in Arabic means heart and pronounced **(qalb)** if the letter **ق** (**qāf**) is mispronounced as **ك** **(kāf)** the word will be **"كلب"** whitch means dog. This means that mispronouncing one letter could change the whole meaning of the word, potentially causing confusion or misunderstanding in communication. This highlights how important accurate pronunciation is in Arabic.

Currently, feedback of accurate pronunciation is mostly provided by human instructors, and it's not always accessible, consistent, or affordable for all learners. As a result, many learners lack effective, immediate support to help them identify and improve their pronunciation errors[9].

To address this issue, the proposed system targets Arabic-speaking children at early stages of language development by analyzing spoken Arabic to detect mispronounced letters and provide immediate, personalized feedback. Using artificial intelligence and speech processing techniques, the project aims to offer an accessible solution that enables users to practice and improve their Arabic pronunciation independently, without the need for constant instructor supervision.

1.2 The Solution

This project proposes an AI-driven pronunciation improvement platform for Arabic-speaking children designed to address common Arabic pronunciation difficulties faced by Arabic-speaking children at early stages of language. These difficulties include articulating challenging Arabic letters, distinguishing between phonetically similar sounds, and producing exact pronunciation without continuous guidance.

The application begins with a placement test to assess the user's initial pronunciation level. Based on the results, users are provided with targeted pronunciation activities that match their proficiency. Users read selected Arabic words, and their speech is recorded and analyzed using speech processing techniques to evaluate pronunciation accuracy.

The platform supports pronunciation improvement through automated assessment, targeted feedback, and structured training. It highlights pronunciation weaknesses, offers personalized practice, and tracks user progress over time to support gradual improvement.

By combining automated evaluation, personalized practice, and structured progression, Faseh (فصيح) delivers an effective and accessible solution for improving Arabic pronunciation.

1.3 Product

1.3.1 Product Vision

Product Vision: **For** Arabic-speaking children **Who** struggle with accurate Arabic pronunciation. **The** Faseh (فصيح) is a mobile AI-powered application for improving Arabic pronunciation.

That evaluates users' pronunciation, identifies errors, and provides tailored feedback to improve speaking accuracy.

Unlike existing language learning platforms and speech assessment systems that are primarily optimized for the English language (e.g. Duolingo).

Our product offers Arabic-specific pronunciation analysis with phoneme-level feedback and personalized practice using artificial intelligence

1.3.2 Product Roadmap

A product roadmap is "a description of the incremental nature of how a product will be built and delivered over time, along with the important factors that drive each individual release. Useful when developing a product that will have more than one release."

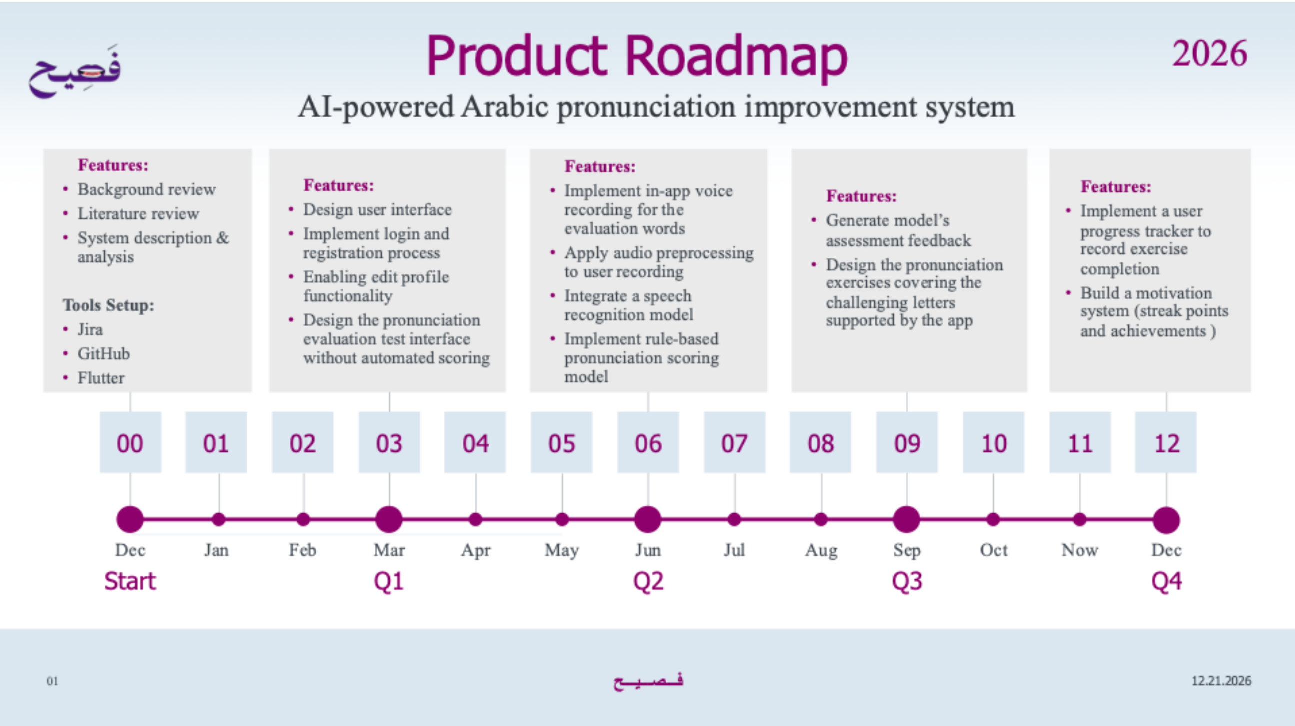


Figure 1:Product Roadmap

1.3.3 Objectives

Product (customer focus-value): why are we building this product? What problems will it solve? Who will benefit (the users), and how?

- **Provide accurate Arabic pronunciation feedback**

Enable users to receive immediate, AI-based feedback on their Arabic pronunciation to help them enhance and improve their pronunciation accuracy.

- **Support children's Arabic pronunciation development**

Help Arabic-speaking children improve their pronunciation of Arabic letters and words.

- **Improve pronunciation confidence**

Help users gain confidence in speaking Arabic by enabling private, self-paced practice with AI-based feedback, without the pressure of speaking in front of a teacher or other learners.

- **Deliver personalized learning experience**

Adapt feedback based on user performance by identifying pronunciation weaknesses and offering targeted practice exercises designed to reinforce correct pronunciation.

- **Provide an easy-to-use mobile interface**

Ensure the application is simple, intuitive, and accessible for users with different technical backgrounds.

- **Encourage continuous practice**

Motivate users to practice regularly through repeated exercises and a progress tracking system that visualizes performance history, score improvement, and practice consistency over time by using streak system.

Project (solution focus-plan): what we need to do to complete the project? what are the steps/stages?

- **Understand user needs and system requirements**

Design and conduct a survey to identify user needs and gain insights that support feature development. This step will be complemented by interviews to obtain deeper qualitative insights.

- **Design the system architecture**

Design a complete system architecture including mobile frontend, backend services, AI model integration, and cloud storage.

- **Develop a mobile application**

Implement the frontend as an Android-based application with integrated audio recording and user interaction features.

- **Develop backend services and APIs**

Build backend services to handle audio uploads, model inference requests, and result delivery to the mobile application.

- **Integrate speech recognition models**

Integrate pre-trained Arabic-capable speech recognition models to transcribe and analyze user pronunciation.

- **Develop a pronunciation scoring model**

Develop a pronunciation scoring model that evaluates users' Arabic pronunciation, generates pronunciation scores, and provides feedback based on pronunciation performance.

- **Test and evaluate the system**

Perform functional testing, usability testing, and performance evaluation to ensure system reliability and accuracy.

Learning (student focus): what will the student learn from completing this project. Something new, something not learned as part of the IT curriculum.

- Gain experience with speech processing and AI models, especially for Arabic language applications.
- Learn how to integrate pre-trained Transformer-based speech processing models (e.g. Wav2Vec2) into real-world systems for speech processing and understanding.
- Develop advanced mobile application development skills using Flutter and Dart.
- Understand audio preprocessing for speech applications.
- Work with cloud services such as Firebase for storage.
- Improve problem-solving and research skills by working on a real AI-based solution.
- Enhance teamwork, project management, and documentation skills through structured sprint-based development.
- Gain experience in testing AI systems and evaluating model performance.

1.3.4 Scope

The Faseh (فصيح) application will be developed as an Android mobile AI-powered application designed to support Arabic pronunciation improvement for Arabic-speaking children using AI-based speech analysis techniques. The application will operate in Modern Standard Arabic (MSA), following correct pronunciation rules appropriate for early-stage language learners.

The scope of the application is limited to automated pronunciation assessment and feedback through user voice recordings. The system will present age-appropriate predefined words for reading, evaluate pronunciation accuracy, and provide corrective feedback and scoring. The application focuses on a selected set of commonly challenging Arabic letters for children, specifically (ق، خ، غ، ص، س، ض), which are initially assessed and then targeted through the provided exercises.

The application will not include full speech conversations, translation features, or support for non-Arabic languages in the initial release. It is intended for general educational pronunciation learning only and does not target individuals with speech disorders or clinical speech impairments, nor does it provide speech therapy or medical diagnosis. Additionally, the application does not cover advanced pronunciation aspects, including control of speech movements, voice tone, speech rhythm, and dialect differences, as these features are beyond the scope of the initial version.

The primary users of the application are Arabic-speaking children at early stages of language development who seek to improve their pronunciation through independent practice and AI-guided feedback. Parents are considered secondary users, with their role limited to account management and supervising their children's learning progress.

1.3.5 Hardware/Software Tools and Cost

Hardware Tools	
Name and Description	Cost *mention date it will be available*
Laptop	Already owned
Mobile device	Already owned
Software Tools	
Name and Description	Cost
Visual Studio (IDE for Coding)	Free
Wav2vec model	Free
Firebase storage service	Free/limited use

Table 1:Hardware/Software Tools and Cost

1.4 Scrum Team

1.4.1 Skill Set Requirements

Technical Skill Required	**What is the current level of the team (*beginner-intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary*) Learning plan**
Flutter & Dart development	Beginner. Skills will be improved through official documentation, online tutorials, and hands-on implementation within the project.
Mobile UI/UX design.	Intermediate.

	Skills will be strengthened by applying established design guidelines, analyzing user feedback, and iterating on wireframes and prototypes to improve usability and consistency.
Scoring model development	Beginner. Skills will be improved by understanding basic pronunciation scoring concepts, implementing simple scoring logic based on model outputs, and refining the scoring approach through testing and evaluation during the project.
Database and cloud management	Intermediate. Skills will be strengthened by managing cloud-based databases and storage services, organizing user data and audio files, and integrating the application with cloud platforms through hands-on practice during development.
Pretrained speech model integration	Beginners. The team has experience working with ML models but limited exposure to integrating pretrained models. The gap will be addressed by reviewing relevant documentation, following guided tutorials, learning API usage and inference workflows, and applying this knowledge directly during the backend development and testing phase.

Version Control (Git/GitHub)	Intermediate. The team has experience working with GitHub and will enhance their skills by reviewing more GitHub documentation, learning advanced features such as branching strategies and conflict resolution, and applying these practices consistently during team collaboration.
API Integration	Intermediate. The team has prior experience working with APIs. Any gaps related to integrating new or more complex APIs will be addressed by reviewing official documentation, testing endpoints in isolated prototypes, and applying best practices during implementation.
Data Handling & Dataset Preparation	Intermediate. The team has experience with basic data preprocessing techniques. To address gaps related to audio-specific data handling, the team will study and apply audio preprocessing methods such as noise reduction and feature extraction before preparing the dataset for model evaluation.
Agile methodology	Intermediate. The team will follow Agile practices by conducting sprint planning, holding regular

	stand-up meetings, and managing tasks and progress using Jira.

Table 2: Skill Set Requirements

1.4.1.1 Learning

This is a new section in the report, explain here what steps have been undertaken to achieve the learning required, and the current stage of your learning.

To achieve the required learning outcomes for this project, several preparatory and exploratory activities have been undertaken. The team is actively engaging with experts who have experience in Natural Language Processing (NLP) and artificial intelligence applications for the Arabic language. Meetings are being arranged with these specialists to gain domain-specific insights, understand common challenges in Arabic speech processing, and apply expert guidance to the project design and implementation.

In parallel, the team is communicating with multiple researchers to identify and evaluate suitable Arabic speech datasets that align with the project requirements. This includes assessing dataset quality, linguistic coverage, and relevance to pronunciation analysis tasks.

Additionally, the team is exploring structured learning resources and online courses to develop proficiency in Flutter programming for mobile application development. These learning efforts aim to support the implementation phase of the project and ensure effective integration between the AI components and the mobile interface.

1.4.2 Roles and Responsibilities

Scrum Team	
Product Owner:	Dr. Reem Alqifari
Developers:	Walah Alsaeed Norah Alkathiri Joud Alzahrani Tala Alsheail
Scrum Master (SM):	Dr. Reem Alqifari

Stakeholders:	GP Examiners

Table 3:Roles and Responsibilities

2 Background

2.1 Introduction

This chapter provides a comprehensive overview of the domain knowledge necessary to fully understand the project. It examines the linguistic characteristics of Arabic pronunciation, with particular emphasis on its phonetic and phonological complexity. In addition, it highlights the distinctive features of Arabic that may pose challenges for Arabic-speaking children during the early stages of language development. To further clarify the nature of the problem addressed in this work, the chapter discusses common pronunciation difficulties, including articulatory errors and inconsistencies in vowel length and emphasis.

In addition, the chapter examines the role of technology in supporting pronunciation learning, with a particular focus on speech processing and artificial intelligence techniques used for pronunciation analysis and feedback. By reviewing both the problem domain (Arabic pronunciation challenges) and the solution domain (technology-driven pronunciation assessment and improvement), this background establishes the conceptual framework required to understand the system design and objectives. The content is presented in an accessible manner, ensuring that readers without prior knowledge of Arabic linguistics or speech processing can fully understand the motivation, relevance, and scope of the proposed solution.

2.2 Arabic Pronunciation and Phonetic Characteristics

Arabic pronunciation refers to the correct articulation of sounds used in spoken Arabic. The Arabic language contains a set of phonetic characteristics that could be unfamiliar to new learners and make pronunciation learning particularly challenging for Arabic-speaking children in early language learning stages.

One important concept in Arabic pronunciation is phonemes, which are the smallest units of sound that distinguish meaning between words. In Arabic, changing a single phoneme can result in a completely different word with a different meaning. This makes accurate pronunciation essential for clear communication[9].

Arabic also includes emphatic letters, which are consonants pronounced with extra emphasis in the vocal tract. These sounds are uncommon in many languages, making them difficult for learners to pronounce accurately. Examples of emphatic letters include /ص/ (ṣād), /ض/ (ḍād), /ط/ (ṭāʾ), and /ظ/ (ẓāʾ). Another key feature is deep throat sounds, which are produced in the back of the throat and require precise control of speech organs (e.g., /ع/ (ʿayn) and /ح/ (ḥāʾ))[7].

In addition, Arabic pronunciation relies heavily on vowel length, meaning that the duration of vowel sounds (short or long) affects word meaning[9]. Mispronouncing vowel length can completely alter interpretation, even when all consonants are produced correctly. For example, /عَلَم/ (ʿalam – flag) differs from /عَالَم/ (ʿālam – world), where the only distinction lies in vowel length, yet the meaning changes entirely.

Due to these characteristics, Arabic pronunciation requires careful attention to sound production, making it one of the most complex aspects of Arabic language learning for individuals without prior exposure to the language.

2.3 Pronunciation Learning Challenges

The development of accurate pronunciation depends on learners' ability to identify and correct their speech errors. For many learners, especially Arabic-speaking children at early stages of pronunciation development, identifying pronunciation mistakes independently is difficult. This challenge is primarily due to unfamiliar sounds when learning a new language.

Traditionally, pronunciation learning depends on feedback provided by human instructors, who guide learners by correcting mispronunciations and modeling correct sounds. However, instructor-based feedback may not always be available, consistent, or scalable, particularly for learners practicing independently. Without immediate feedback, learners may repeat incorrect pronunciation patterns, slowing progress and reducing speaking confidence [9].

To address these challenges, modern pronunciation learning approaches increasingly rely on technology-based systems that provide automated, immediate, and consistent feedback, enabling learners to practice independently while receiving structured guidance.

2.4 Role of Technology in Pronunciation Learning

Technology has become an important tool in modern language learning. Digital learning platforms help learners practice language skills on their own while getting structured guidance[10]. In pronunciation learning, technology can analyze spoken input automatically and compare it to correct pronunciation standards [11].

Artificial intelligence helps systems process speech data, find pronunciation weaknesses, and give feedback that matches the needs of each learner. This supports personalized learning experiences and lets learners practice at their own pace without needing constant supervision [12].

2.5 Speech Processing and Automated Pronunciation Assessment

Speech processing refers to the analysis of spoken audio signals to extract linguistic information. In applications that focus on pronunciation, speech processing techniques are applied to examine recorded speech and evaluate how closely it aligns with correct pronunciation patterns[13].

Automated pronunciation assessment involves the use of computational methods to evaluate pronunciation accuracy without human involvement **[**13**]**. These systems can produce pronunciation scores and feedback based on detected pronunciation weaknesses**,** ensuring objective and consistent assessment. When combined with artificial intelligence, automated evaluation systems support structured speech learning and enable progress tracking over time.

2.6 Supporting Tools

The proposed system is developed to fill the gap in Arabic pronunciation learning technology. It is implemented as a mobile-based application that uses commonly available hardware, such as smartphones with built-in microphones, to record user speech. The system does not require any specialized or external hardware. On the software side, artificial intelligence and speech processing technologies are used to analyze audio input and provide pronunciation feedback.

3 Literature Review

This section presents a descriptive overview of existing research and systems related to Arabic language learning, with a particular focus on pronunciation assessment and technology-assisted educational platforms. The purpose of this literature review is to analyze relevant studies to identify current approaches, limitations, and opportunities within this domain. By examining similar and related work, this section helps establish the requirements of the proposed project and positions it in relation to existing efforts.

Arabic Speech Mispronunciation Detection Dataset [14]

A primary obstacle to the development of automatic pronunciation assessment systems is the scarcity of large, well-annotated corpora that accurately reflect authentic mispronunciation patterns. To address this limitation, a dataset tailored for Arabic speech mispronunciation detection was introduced. It comprises audio recordings of the most frequently used Arabic words, spoken by children and annotated for segmental errors.

The dataset features recordings of the 100 most common words, produced by 100 Egyptian children aged 2 to 8 years, with expert listeners identifying phoneme-level pronunciation deviations. Such a dataset is particularly valuable because children's mispronunciations represent naturally occurring pronunciation errors, rather than synthetic or artificially generated mistakes, thereby offering a realistic resource for training and evaluating mispronunciation detection models.

The primary contribution of this dataset is its emphasis on segmental pronunciation errors, enabling researchers to develop and benchmark models that detect pronunciation faults at a detailed level, which is essential for providing targeted pronunciation feedback. Although the dataset was collected from children and focuses on the Egyptian dialect, it underscores the significance of annotated mispronunciation resources for constructing accurate speech evaluation systems. Notably, the study did not evaluate specific models or integrate the dataset into a pronunciation learning platform. This omission highlights the ongoing need for systems that combine mispronunciation detection with learner-oriented feedback and training mechanisms. The proposed project addresses this gap by integrating mispronunciation evaluation with personalized guidance and progress tracking.

Arabic pronunciation assessment for Saudi Arabian Student: Corpus development and system architecture [15]

Research published from King Fahd University of Petroleum & Minerals (2024) that explains the process of building the largest speech corpus in Saudi Arabia designed for children from the ages of 6-15 years. This corpus was developed to assist the children in their pronunciation to ultimately improve their reading ability, since any deficiency in it could negatively affect a student's progress in other subject areas according to the researchers. This study demonstrates the feasibility of large-scale Arabic pronunciation assessment systems in providing young Arab learners with a valuable resource for mastering their native tongue.

The researchers develop an interactive Arabic reading tutor using a machine-learning/deep-learning web-based platform. The assessment process begins by asking the student to read some text that is displayed on the screen, originating from an article/lesson, a paragraph, a sentence, a word or a letter. Following this, the reading assessment system evaluates the student's reading, detecting any mistakes or errors made (e.g., spending more than average time to read, prosodic mistakes, or mispronunciation in general). In the case of a mistake, the intelligent tutor system will be engaged, where various activities are presented to address the student's reading issues.

In conclusion, after the researchers have successfully developed and validated their received recordings, it was used to develop an automatic Pronunciation Assessment system, which leverages technologies to provide effective feedback on pronunciation.

While this work focuses on Arabic reading and pronunciation evaluation, it does not target independent learners or provide fine-grained pronunciation training for adult or non-native Arabic learners, which highlights an opportunity for further research.

Arabic Mispronunciation Recognition System Using LSTM Network [8]

A study published in 2023, showcasing the process of building a model to resolve the Arabic pronunciation challenges experienced by speakers, due to the presence of phonemes with complex places of articulation and acoustic characteristics that differ from those found in many other languages. Mispronunciation of certain Arabic sounds may lead to changes in word meaning and negatively affect communication, making accurate pronunciation a critical component of Arabic language learning.

A study proposed an automatic Arabic mispronunciation recognition system using speech processing and deep learning techniques. The research focused on detecting mispronounced Arabic letters by analysing spoken speech signals.

The system employed Mel-Frequency Cepstral Coefficients (MFCCs), which are features extracted from speech signals to represent the spectral characteristics of sound in a way that reflects human auditory perception [16], for feature extraction , and a Long Short-Term Memory (LSTM) neural network, a type of recurrent neural network capable of learning long-term dependencies in sequential data such as speech, for classification [17]. The results demonstrated that deep learning approaches are capable of effectively identifying pronunciation errors at the phoneme level, particularly for Arabic sounds with complex articulation properties.

The findings of the study highlight the feasibility of using artificial intelligence to support Arabic pronunciation assessment. Phoneme-level analysis was shown to be effective in detecting subtle pronunciation differences that are difficult to identify through traditional learning methods. However, the proposed system primarily focused on mispronunciation detection and classification accuracy, without incorporating learner interaction, personalized feedback, pronunciation training activities, or progress tracking mechanisms.

A novel framework for mispronunciation detection of Arabic phonemes using audio-oriented transformer models [18]

A recent study proposed a novel framework for detecting mispronunciations of Arabic phonemes using audio-oriented transformer models (transformer-based models designed to learn representations directly from raw speech signals). The study focused on evaluating pronunciation at the phoneme level (individual sound units rather than full words or sentences), addressing challenges related to fine-grained phonetic distinctions in Arabic. The proposed framework evaluated several state-of-the-art transformer-based speech models, including Wav2Vec, HuBERT, SEW, and UniSpeech, using a dataset consisting of Arabic phoneme pronunciations produced by multiple speakers. The results showed that transformer-based models, particularly UniSpeech, achieved higher accuracy in phoneme-level mispronunciation detection compared to earlier deep learning approaches, while also supporting speaker-independent evaluation (model robustness across different speakers without speaker-specific adaptation)[8].

While the study successfully demonstrated the effectiveness of transformer models in detecting Arabic phoneme mispronunciations, it primarily focused on classification performance and system accuracy. It did not address learner interaction, personalized pronunciation feedback, or structured pronunciation training. This highlights a gap between accurate mispronunciation detection and practical pronunciation improvement for learners, which the proposed project aims to address.

Arabic ASR on the SADA Large-Scale Arabic Speech Corpus with Transformer-Based Models [19]

A recent study introduced a multilingual Arabic pronunciation evaluation dataset and proposed an improved framework for mispronunciation detection utilizing transformer-based speech models. The research sought to advance automatic analysis of Arabic phoneme pronunciation by incorporating a wide range of dialectal and phonetic variations, thereby addressing the limitations of traditional speech recognition systems that struggle with isolated phonemes and speech variability.

The study utilized an existing Arabic speech dataset of spoken Arabic phonemes to capture diverse real-world pronunciation patterns. Several transformer-based speech models, including Wav2Vec2, and other attention-based architectures, were evaluated for their ability to classify correct and incorrect phoneme pronunciations. The results demonstrated that transformer models pretrained on extensive speech corpora outperformed conventional methods, especially in managing pronunciation inconsistencies and noise present in spontaneous speech.

The study further demonstrated that large-scale pretrained speech models are effective not only for word- and sentence-level transcription but also for fine-grained phoneme discrimination when properly adapted. These findings indicate that models such as Wav2Vec2 can serve as robust foundations for pronunciation evaluation tasks, ensuring both accuracy and resilience across diverse speaker variations.

Although the study established the technical feasibility of transformer-based pronunciation detection, it did not examine methods for delivering personalized feedback or structuring pronunciation training for effective learning. This gap between automated mispronunciation detection and actionable learner feedback will be addressed in the proposed project by integrating advanced speech models with tailored feedback and practice mechanisms within an interactive learning platform.

An Approach for Pronunciation Classification of Classical Arabic Phonemes Using Deep Learning [9]

A recent study examined integrating advanced speech signal features and machine learning techniques to enhance Arabic pronunciation assessment and mispronunciation detection. The research focused on extracting multiple acoustic features and comparing various classification algorithms to evaluate Arabic phoneme pronunciation. The study combined spectral coefficients and temporal characteristics to capture articulation pattern variations, which are essential for distinguishing between correct and incorrect pronunciations in spoken Arabic.

The experimental results indicated that integrating diverse speech features with appropriate classifiers significantly enhanced the detection of mispronounced phonemes compared to single-feature methods. This finding underscores the importance of feature engineering in pronunciation evaluation, as it facilitates more precise modeling of speech variability and phonetic distinctions. However, the study did not address the development of an interactive feedback system or a learning platform, thereby leaving a gap between technical evaluation methodologies and practical end-user applications.

These findings support the proposed project's direction by demonstrating that robust feature extraction and classification techniques can improve the accuracy of pronunciation assessment. Building on these insights, Faseh (فصيح) seeks to integrate advanced speech-feature processing with personalized feedback and structured training to deliver a learner-centered pronunciation-improvement experience.

Towards stable AI systems for Evaluating Arabic Pronunciations [20]

A study in 2025 that aimed to improve the performance of common Modern Arabic ASR systems such as wav2vec 2.0, which excels at word and sentence level transcription, but lack in classifying isolated letters. This study sheds light on the lack of phenome-level classification in the wav2vec model, where it was needed to apply additional padding or context around the targeted phoneme to achieve reasonable accuracy.

To develop higher accuracy, the researchers have begun by collecting their data through a web-based and mobile-based platform, where users must submit an audio recording including a specific single letter, which was annotated by linguistic experts to ensure consistency. After developing a corpus, the researchers fine-tuned a wav2vec 2.0 model and a lightweight MLP baseline to achieve a higher accuracy in phenome-level classification. This method showed great improvement. For example, it reduced confusion between phonetically similar letters, which exist commonly in the Arabic language (e.g. ق vs. ك).

This study highlights the limitations of currently existing Modern Arabic ASR systems that have been shown to struggle with phenome-level recognition, which is essential for language learning application. The proposed approach that is clarified in this study improves isolated letter classification and improves accuracy and robustness, but it primarily focuses on performance rather than user-centered pronunciation assistance.

Improving Mispronunciation Detection of Arabic Words for Non-Native Learners Using Deep Convolutional Neural Network Features [21]

This paper, published in 2020, discusses the process of building a model that correctly identifies mispronunciation of Arabic words within a computer assisted language learning (CALL) system, providing a stress-free language learning environment, which enables students to learn at their own pace. The holy book of Islam is the Quran, proper recitation of Quran requires the reader to follow specific rules, known as tajweed. To develop the suggested CALL system supporting proper Arabic pronunciation, the researchers have begun by collecting a dataset involving recitations of the Quran from native and non-native Arabic speakers, then a spectrogram (2D visual representation of an audio signal's frequency spectrum) was constructed from the speaker's audio, these spectrograms has been passed onto a CNN model in order to identify the most important features for Arabic mispronunciation detection.

This study strongly demonstrates the effectiveness of using deep convolutional neural network (CNN) in comparison with traditional machine learning approaches that depend on handcrafted features (e.g. MFCC) and classifiers trained on manually extracted acoustic representations. This approach focuses entirely on building a word-level model for non-native users that determines if a word was pronounced correctly or not but does not offer any personalized feedback yet. In contrast, our proposed Faseh (فصيح) system is targeted toward both native and non-native users and aims to go beyond model development by integrating pronunciation scoring and personalized feedback.

After conducting a comparative analysis of all the reviewed studies mentioned above, the results showed 3 major research directions in Arabic pronunciation assessment technology: phoneme-level mispronunciation detection, speech corpus development, and model centered performance evaluation, where the model is assessed using metrics computed on the developed corpus. Several studies, such as those using LSTM networks, transformer-based models, and wav2vec2 fine-tuning, mainly focus on phoneme-level detection for native and non-native Arabic speakers. While these different approaches contribute to improving fine-grained pronunciation assessment, they are highly model-driven and focus on its classification accuracy and performance rather than user's interaction. Other works concentrate on dataset development and system architecture, including the large-scale Saudi children's speech corpus and specialized mispronunciation datasets for other dialects. However, these studies emphasize data collection and management over designing interactive learning systems. Lastly, some research targeted word-level mispronunciation detection using neural network techniques to offer some simple pronunciation scoring within a CALL framework (usually targeting non-native speakers), but these frameworks typically work as standalone assessment tools, offering no structured feedback, progress monitoring, or personalized learning exercises.

In contrast, our proposed system Faseh (فصيح) is designed to be more learner-centered, combining automatic word-level pronunciation assessment with structured personalized feedback, improvement exercises, and progress tracking. Which is unlike purely model-based or dataset focused studies, therefore addressing the gap between technical mispronunciation detection and a complete learning experience.

3.1 Dataset Exploration and Selection

This section reviews the speech datasets that were considered for use in the proposed Arabic pronunciation assessment project. Each dataset is discussed in terms of its content, relevance to the project objectives, and the reasons for its acceptance or exclusion. The goal of this evaluation was to identify a dataset that best supports phoneme-level pronunciation analysis for Arabic-speaking children. A comparative summary of the datasets is provided at the end of the section.

SADA 2022 Arabic Speech Dataset (Kaggle) [22]

Dataset Description

The Saudi Audio Dataset for Arabic (SADA) is a large-scale Arabic speech corpus published by SDAIA in collaboration with the Saudi Broadcasting Authority. It contains approximately 667 hours of Arabic audio collected from over 57 television programs, mainly in Saudi dialects. The dataset is split into training, validation, and testing sets (418 hours for training and 10 hours each for validation and testing) and includes 4,563 audio files with an average duration of about 10 minutes.

Why It Was Considered

SADA was considered due to its large size, high-quality audio, and strong representation of Saudi Arabic speech, making it suitable for speech-based machine learning tasks.

Why It Was Not Used

Despite its advantages, SADA is designed for general speech recognition and does not include annotations for pronunciation errors or phoneme-level assessment. In addition, the data is sourced from broadcast media rather than learner speech and is not tailored for children or early-stage learners. Therefore, it does not meet the specific requirements of this project.

Arabic Pronunciation Assessment Dataset (MDPI, 2023)[8]

Dataset Description

This dataset was created due to the lack of a standard Arabic pronunciation corpus and was collected through two controlled sessions. The training set includes 7,800 utterances from 30 native Arabic speakers, while the testing set contains 3,120 utterances from 11 native and 42 non-native speakers aged 9–65 from 17 countries. Recordings were captured at 44.1 kHz with noise reduction applied. The dataset is speaker-independent, text-dependent, and annotated using binary labels (correct vs. incorrect pronunciation).

Why It Was Considered

The dataset directly targets Arabic pronunciation assessment and includes labeled mispronunciation data from both native and non-native speakers and controlled audio recordings.

Why It Was Not Used

The dataset is not publicly available for reuse, and it is also not specifically designed for child-oriented pronunciation learning.

Arabic Pronunciation Assessment Corpus for Saudi Primary Students (Springer)[15]
Dataset Description

This dataset is a speech corpus developed to support Arabic pronunciation assessment for primary school students in Saudi Arabia. It includes audio recordings from more than 503 participants aged between 6 and 15 years, resulting in approximately 33 hours of recorded speech. The recordings follow Modern Standard Arabic pronunciation rules and focus on children’s oral reading performance. The corpus was designed specifically for educational pronunciation evaluation and includes teacher-reviewed data to ensure pronunciation accuracy and linguistic validity.

Why It Was Not Used

Although this dataset has not yet been used in the current implementation phase due to pending access approval, it is considered the most suitable dataset for the project. This is because it was specifically developed for Arabic pronunciation assessment for Saudi primary school students and follows Modern Standard Arabic pronunciation rules, which closely align with the target users of the proposed system. In addition, the dataset contains age-appropriate Saudi children's speech recordings, making it more representative of the project's intended learning environment compared to other available datasets that focus on different dialects, such as Egyptian Arabic. These characteristics are expected to support more accurate pronunciation assessment and better model generalization for the target user group once the dataset becomes available for integration.

Arabic Speech Mispronunciation Dataset for Egyptian Children[13]

Dataset Description

This dataset contains annotated recordings of Arabic mispronunciation errors in isolated words from 100 Egyptian children aged 2–8 years. It covers the 100 most frequently used Arabic words, with each child pronouncing 50 or 100 words, organized into separate folders. Recordings were captured in mono at a 44.1 kHz sampling rate using Audacity, with noise reduction applied, and each word was labeled as correctly or incorrectly pronounced.

Why It Was Considered

The dataset was considered because it provides labeled mispronunciation data from children and focuses on pronunciation error detection, which is a core objective of the project.

Why It Was Used

Despite its limitations, the dataset is currently being used during the early experimentation phase for preliminary model training and testing. Its annotated child speech recordings provide useful examples of pronunciation errors that support initial pronunciation analysis and evaluation workflows. However, because the dataset focuses on isolated words, binary pronunciation labels, and Egyptian dialect speech, it does not fully align with the project's target users or Modern Standard Arabic pronunciation requirements. As a result, relying on this dataset alone may limit the overall pronunciation assessment accuracy and generalization of the final system.

Dataset Comparison Summary

Dataset	Main focus	Age group	Dialect	Pronunciation Error Labels	Size	Public Availability
SADA [22]	General Arabic speech / ASR	9-65	Saudi dialects + other dialects	No	667 hours	Yes
Arabic Pronunciation Assessment Dataset [8]	Pronunciation correctness	9–65 years	Mixed Arabic	Yes	10,920 utterances	No
Arabic Pronunciation Assessment Corpus for Saudi Primary Students [15]	Pronunciation & reading assessment	6–15 years	Modern Standard Arabic	Yes	33 hours	Limited (Request based)
Arabic Speech Mispronunciation Dataset (Egyptian Children) [14]	Mispronunciation detection (isolated words)	2–8 years	Egyptian	Yes	100 children × 100 words	Yes

Table 4: Dataset Comparison

3.2 Competitive Product Analysis

This section analyzes four Arabic language learning applications to evaluate their features and instructional approaches. The analysis aims to identify strengths, limitations, and existing gaps, particularly in supporting pronunciation learning, to justify the need for the proposed system.

- **Duolingo**

  Duolingo is one of the most popular language learning platforms worldwide, offering courses in multiple languages through a gamified mobile and web-based experience. The platform focuses on vocabulary acquisition, basic grammar, reading, listening, and limited speaking exercises.

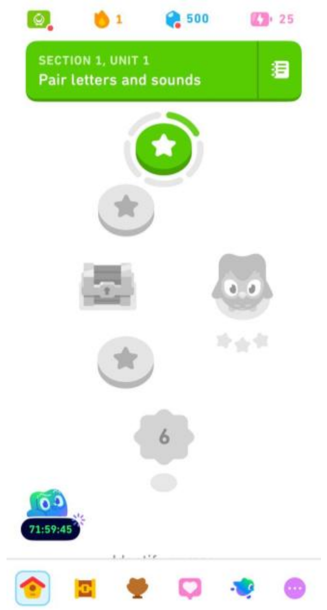


Figure 4:Duolingo Lesson Path

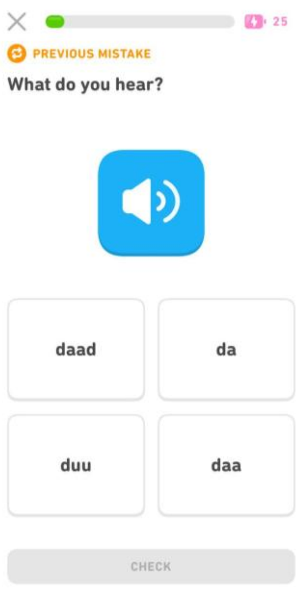


Figure 2:Duolingo Listening Exercise

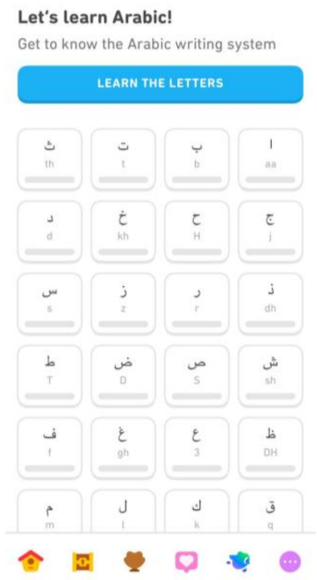


Figure 3:Duolingo Arabic Alphabet Learning

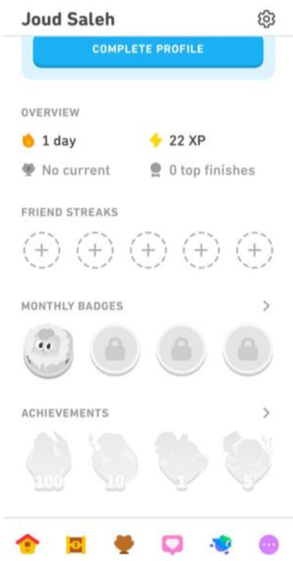


Figure 7:Duolingo User Profile

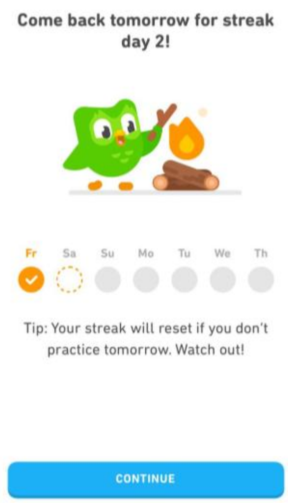


Figure 5:Duolingo Streak Reminder

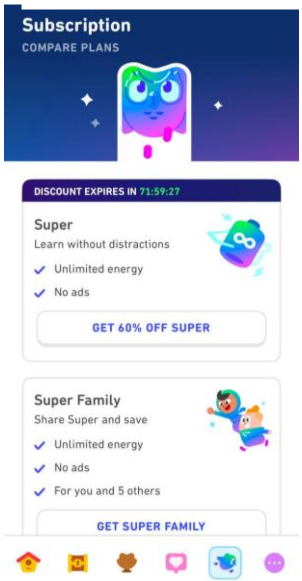


Figure 6:Duolingo Subscription Plans

Main Features:

- Gamified learning experience: Incorporates progress levels, streak tracking, points, and rewards to encourage consistent learning **(See Figure 7)**, where the streak and XP system are displayed.
- Comprehensive Skill Support: Supports multiple aspects of language learning such as vocabulary, grammar, reading, listening through interactive audio exercises **(See Figure 2)**, where users are asked to identify spoken words.
- User-Friendly Interface: designed with a simple and intuitive layout that supports beginner and early-stage learners **(See Figure 4)**.
- Short and Engaging Lesson Structure: Lessons are structured to be concise while maintaining learner engagement.
- Wide Language Availability: Provides learning support for over 40 languages, including Arabic.

Drawbacks:

- Pronunciation Limitations: Focuses on general language learning rather than targeted pronunciation improvement
- Ad Interruptions: The free version includes frequent advertisements, which may disrupt the learning experience.
- Premium Feature Dependency: Access to advanced features such as offline learning, unlimited attempts, and an ad-free experience is restricted to paid subscription users **(See Figure 6)**.

- **AlifBee**

AlifBee is an Arabic language-learning application designed mainly for non-native learners. It focuses on teaching Modern Standard Arabic (MSA) through a structured curriculum that progresses from beginner to advanced levels, covering reading, writing, listening, and speaking skills.

Figure 10: Alifbee Level Interface

Figure 9: Alifbee Leaderboard Screen

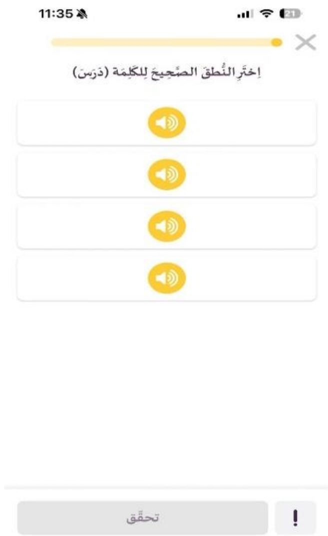


Figure 8: Alifbee Listening Exercise

Figure 12: Alifbee Audio Practice Exercise

Figure 11: Alifbee Audio Practice Exercise

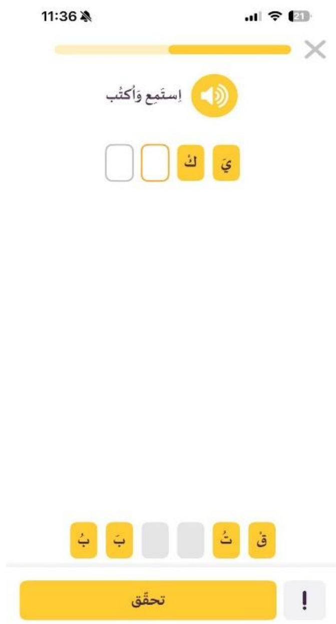


Figure 13: Alifbee Writing Exercise

Main Features:

- Level-Based Learning Path: Organizes learning into structured levels that allow learners to progress gradually, **(See Figure 10)**.

- Gamified Motivation System: Encourages engagement through ranking and competitive elements, **(See Figure 9)**.
- Listening-Based Activities: Provides audio-focused exercises to enhance listening and recognition skills, **(See Figure 8)**.
- Interactive Audio Practice: Allows learners to repeat and practice pronunciation through guided microphone-based tasks, **(See Figures 11 and 12)**.
- Write After Listening Activity: Integrates listening and writing skills by asking learners to write words after hearing them, **(See Figure 13)**.

Drawbacks:

- Limited Pronunciation Focus: Pronunciation practice is included only as a supportive component and lacks detailed analysis.
- No Phoneme-Level Feedback: Learners are not informed about specific letter-level pronunciation mistakes.
- General Language Orientation: The application prioritizes overall language learning rather than targeted pronunciation improvement **(See Figure 10)**.

- **Arabee Family**

  Arabee Family is an educational mobile application that supports Arabic language learning for children through interactive activities. It focuses on early language skills such as listening, speaking, reading, and writing by using games, songs, stories, and visual exercises, while providing a child-friendly environment with parental involvement.

Figure 14 Arabee Family Family Management

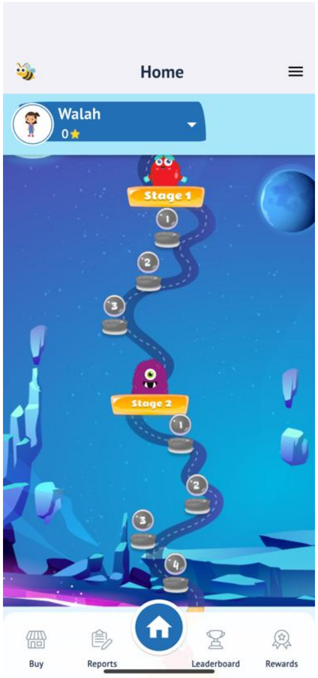


Figure 15 Arabee Family Level Progression

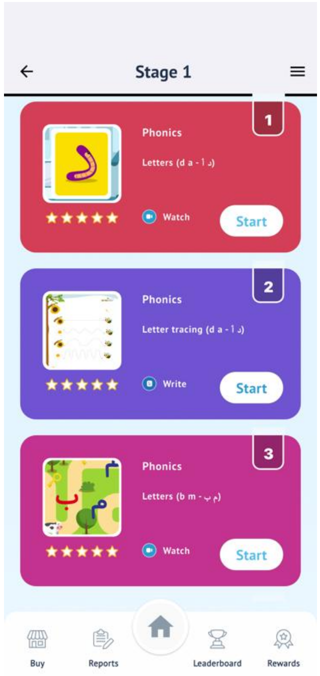


Figure 16 Arabee Family Stage Selection

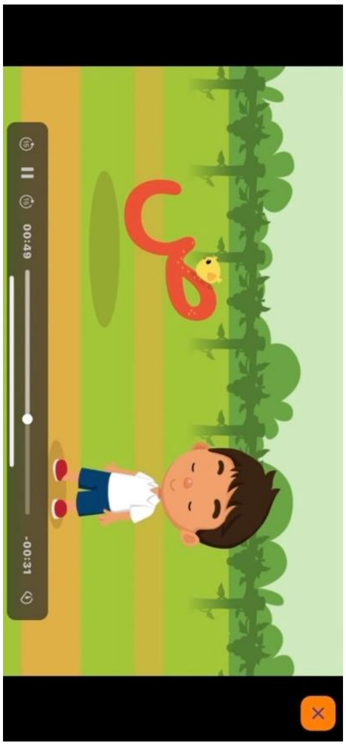


Figure 17 Arabee Family Learning Video

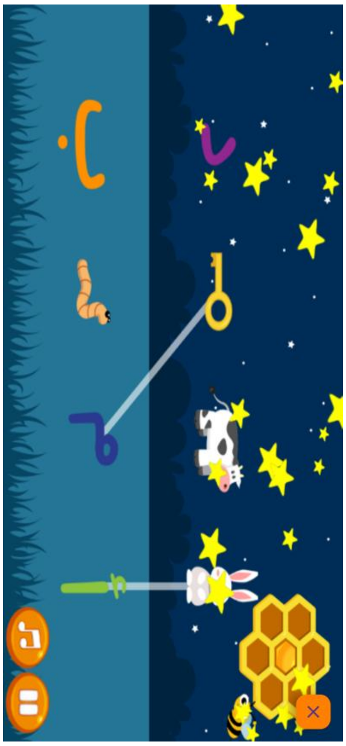


Figure 18 Arabee Family Interactive Game

Figure 19 Arabee Family Letter Tracing Exercise

Main features

- Interactive language activities: The application offers games and interactive exercises that help children practice basic Arabic vocabulary and language concepts in an engaging way **(See Figure 18)**.
- Story-based learning: Animated stories are used to enhance Arabic letters learning and maintain learner engagement **(See Figure 17)**.
- Writing practice through letter tracing: The application provides letter tracing activities that help children develop Arabic writing and letter formation skills **(See Figure 19)**.
- Parental progress tracking: Parents can monitor their children's milestones, including completed lessons and practiced activities, allowing better involvement in the learning process **(See Figure 15)**.
- Gamification and leaderboard system: A leaderboard is used to motivate children by encouraging participation and consistent practice through competition and rewards.

Drawbacks

- General language learning: The application emphasizes overall Arabic learning rather than specifically targeting pronunciation improvement.
- Lack of detailed pronunciation feedback: While listening activities are provided, the system does not analyze or correct pronunciation errors in a personalized manner.
- Limited feature access: Most of the features mentioned above are only restricted to a premium subscription, which might not be affordable for all users who wish to learn Arabic, therefore resulting in limited accessibility.

- **Teach Me Arabic**

  Teach Me Arabic App is an innovative educational platform designed to teach Arabic to non-native speakers, utilizing the latest interactive learning technologies through smartphones. It aims to develop foundational Arabic skills such as reading, writing, or listening through many features.

Figure 20: Teach Me Arabic AI Coach

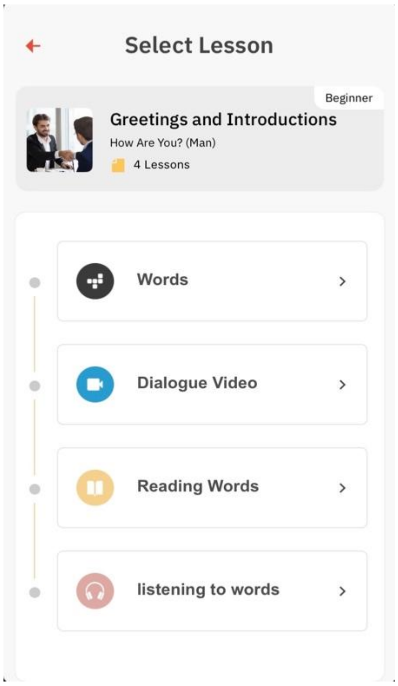


Figure 21: Teach Me Arabic Lesson

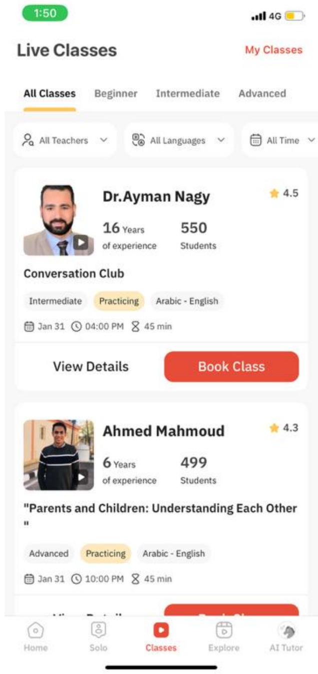


Figure 22: Teach Me Arabic Class Booking

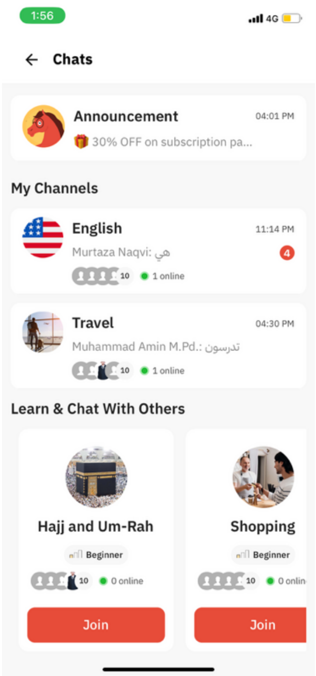


Figure 23: Teach Me Arabic Chat Channels

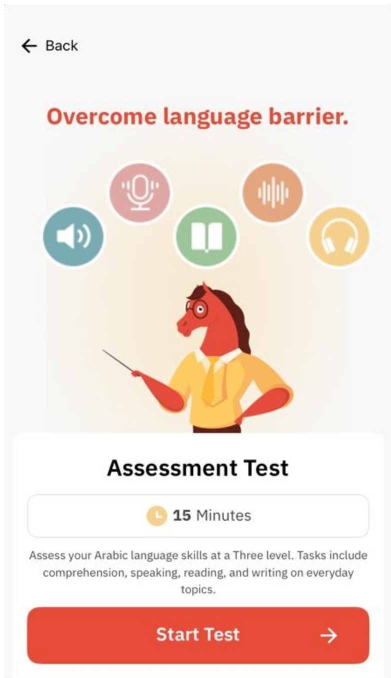


Figure 24: Teach Me Arabic Assessment Test Instructions

Figure 25: Teach Me Arabic Explore Page

Main features

- AI chat coach: An AI-powered chatbot that enables users to chat in Arabic while correcting any typing mistake. The chatbot provides brief correction without clarifying the grammatical rules behind it in detail, then continuing the conversation naturally (e.g. user types "اريد" chatbot will respond with "يجب إضافة همزة فوق الالف") **(See Figure 20)**.
- Level-Based Learning Path: Organizes learning content into sequential levels that help learners progress gradually in Arabic. **(See Figure 21)**.
- Live classes with tutors: Users can book live classes with real-life Arabic teachers who vary in years of experience. These classes have various topics that aim to improve the speaking skill, such as daily conversations or common scenarios (e.g. "What time is it?"). **(See Figure 22)**.
- Chat channels with other app users: The application provides themed chat channels, where users can enter to socialize with other users and improve their conversation skill. These channels are monitored by the AI chat coach mentioned above, they also can vary in topics (e.g. Hajj and Umrah, Shopping, or Travel). **(See Figure 23)**.
- Assessment Arabic test: Users can take a test that determines their proficiency level (beginner, intermediate, or advanced). The test includes tasks that evaluate comprehension, speaking, reading, and writing skills on everyday topics. **(See Figure 24)**.
- Video explore page: The application offers an explore page that presents a collection of educational videos that vary in content, designed to support the development of skills such as speaking, reading, etc. **(See Figure 25)**.

Drawbacks

- General language learning: The application emphasizes overall Arabic learning rather than specifically targeting pronunciation improvement.
- Lack of detailed corrective feedback: The provided feedback in this application is highly superficial, as users are informed that a mistake has occurred, without receiving any grammatical explanation that can assist them to avoid such mistakes in the future and form a sufficient grammatical foundation.
- Limited feature access: Most of the features mentioned above are only restricted to a premium subscription, which might not be affordable for all users who wish to learn Arabic, therefore resulting in limited accessibility.

4 Comparison between similar applications

This section introduces the comparison features used to evaluate the proposed application against selected existing competitors. These features represent key functional and technical capabilities relevant to Arabic learning applications. Each feature is explained to ensure clarity and consistent interpretation before presenting the comparative results.

User account: The ability of the system to provide each learner with a personal profile displaying account details, language level, and learning progress indicators.

Progress tracking: The ability of the application to display the learner's completed exercises and practice history.

Ai based pronunciation assessment: The ability of the application to automatically evaluate a learner's pronunciation using artificial intelligence techniques.

Personalized pronunciation feedback: The ability of the system to generate a personalized exercise plan according to the learner's performance.

Gamification: The ability of the system to support Arabic learning through interactive game-based activities, such as matching and drawing exercises.

Feature	Duolingo	AlifBee	Arabee Family	Teach Me Arabic	فصيح
Arabic language support	✓	✓	✓	✓	✓
Multiple language support	✓	✗	✗	✓	✗
User account	✓	✓	✓	✓	✓
Sub-accounts for children	✗	✗	✓	✗	✓

Progress tracking	✓	✓	✓	✓	✓
Ai based pronunciation assessment	✓	✓	✗	✗	✓
Personalized pronunciation feedback	✗	✓	✗	✗	✓
Gamification	✓	✗	✓	✗	✗

Table 5:Comparison between similar applications

Overall, the comparative analysis shows clear differences in focus and functionality among the reviewed Arabic learning apps. **Duolingo** offers strong gamification, progress tracking, and support for multiple languages. However, its pronunciation features are general and do not provide personalized or phoneme-level feedback. **AlifBee** follows a structured learning path that guides learners through progressive stages of Arabic learning, but its pronunciation support is limited and lacks AI-based assessment or detailed corrective feedback. **Arabee Family** targets young learners with interactive games, stories, and parental tracking, but it focuses on general language skills instead of precise pronunciation analysis and depends heavily on premium access. **Teach Me Arabic** emphasizes communication-oriented learning through chatbots, live classes, and social interaction, but its feedback is basic and does not specifically address pronunciation errors. In contrast, Faseh (فصيح) stands out by combining AI-based pronunciation assessment, personalized feedback, progress tracking, and child-oriented account management into one system. It addresses the main gaps found in other apps and offers a more focused solution for learning Arabic pronunciation.

5 System Requirements

5.1 System Users

The primary users of the Faseh (فصيح) application are Arabic-speaking children in the early stages of language development who seek to improve their pronunciation through independent practice with AI-based feedback. The target age group is 5 to 13 years, corresponding to elementary school students who may require additional support in accurately producing specific Arabic letters and sounds. These users are classified as early-stage Arabic learners or individuals who need assistance with pronunciation refinement. They are expected to possess basic familiarity with smartphones and mobile applications. Children primarily engage with pronunciation exercises, audio recording features, feedback mechanisms, and motivational elements such as streak points.

The secondary users of the system are parents, whose responsibilities are limited to account management and supervision of their children's learning progress. Parents are generally adults aged 25 to 80 years and are expected to have at least a secondary-level education. Their technical expertise typically ranges from basic to intermediate proficiency with smartphones. Within the application, parents create and manage child profiles, monitor individual progress, and update account information as needed. They do not directly participate in pronunciation exercises; instead, they oversee system access and monitor performance tracking to support their child's learning.

5.2 Requirements Elicitation and Analysis

In this section, we present the methods used to for the requirements elicitation process, which includes identifying and understanding the needs, expectations and challenges of stakeholders to define the functional and non-functional requirements of our proposed system.

To gather requirements, a questionnaire was conducted targeting parents and Arabic language teachers of children across different age groups. The aim was to understand pronunciation challenges, learning difficulties, and user expectations for an AI-based pronunciation improvement system. A total of 139 responses were collected.

Additionally, we observed children’s interaction with an existing Arabic learning application (AlifBee), to analyze its strengths and limitations, particularly in relation to pronunciation support and feedback mechanisms.

Method	No. of Questions	No. of Responses
Questionnaire	15	139
Observations	12	8

Table 6: Questionnaire and Observations

5.2.1 Questionnaire

After identifying the primary target users of the application and ensuring a sufficient sample size for meaningful analysis, we designed and distributed an online questionnaire targeting parents and teachers of Arabic-speaking children in early stages of language development. The aim was to collect responses from at least 30 participants to ensure diverse and reliable input regarding pronunciation challenges and educational needs.

To maximize participation, the questionnaire was shared through WhatsApp parent groups, educational communities, and social media platforms. As a result, we successfully collected 139 responses, significantly exceeding the minimum required sample size. This large number of responses enhances the reliability and generalizability of the findings.

The questionnaire was structured based on insights derived from background research, literature review, and analysis of existing educational applications. It included a combination of multiple-choice, Likert-scale, and open-ended questions. The purpose was to explore pronunciation difficulties faced by children, parents' expectations of an AI-based pronunciation system, preferred learning features, and attitudes toward technology-supported learning.

The collected responses provided valuable insights that directly informed the functional and non-functional requirements of the application. A detailed summary of the questionnaire structure and complete response data is provided in Appendix (A).

5.2.2 Demographics of participants

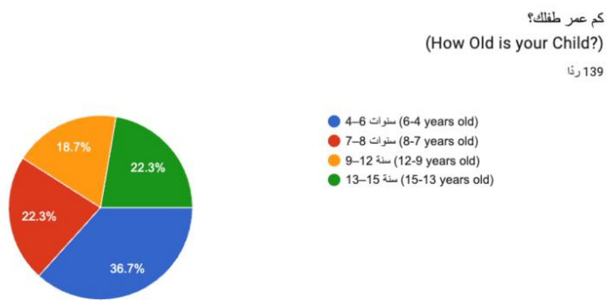
As shown in Figure 26, the questionnaire's initial inquiries were designed to gain a better understanding of the target user's characteristics. Collecting this information helps identify the age groups, number, and the spoken language of the children, providing valuable context for designing system features that better align with the needs of potential users.

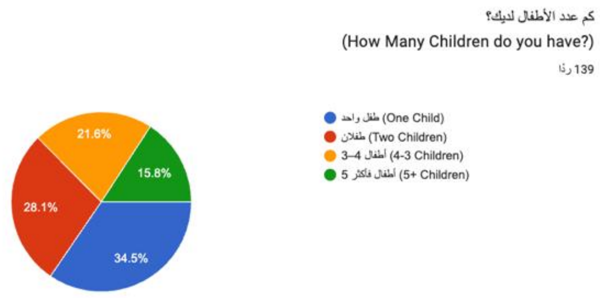
Regarding the age distribution of children, 36.7% were aged 4–6 years, followed by 22.3% aged 7–8 years, 18.7% aged 9–12 years, and 22.3% aged 13–15 years. This indicates that although the primary focus is early-stage learners, pronunciation difficulties extend across a broader age range, suggesting that the system should be adaptable to multiple developmental stages.

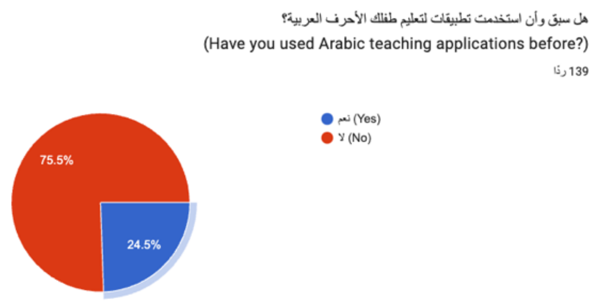
In terms of the number of children per household, 34.5% of respondents had one child, 28.1% had two children, 21.6% had 3–4 children, and 15.8% had five or more. This highlights the importance of supporting multiple child profiles within a single parent account, enabling individualized progress tracking for each child.

Concerning the language spoken at home, 70.5% reported that their children speak Arabic only, while 23% speak mostly Arabic with some English. A small percentage 6.5% reported mostly English usage. Given that most children are Arabic speakers, the system interface and instructions will be provided primarily in Arabic.

When asked about previous use of Arabic learning applications, 75.5% of participants indicated that they had not used any such applications before, while 24.5% had. The applications mentioned included لمسة, زكريا, and others. The high percentage of non-users suggests a strong opportunity to introduce a specialized pronunciation-focused application.







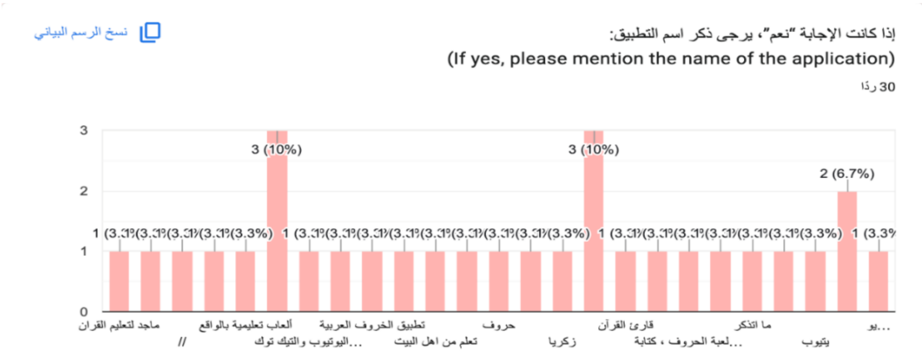


Figure 26: Demographics of participants (Questionnaire)

5.2.3 Questionnaire Results

The questionnaire received 139 responses, providing valuable insights into children's pronunciation challenges and parental expectations for an AI-based Arabic pronunciation learning application. The following results summarize the key findings related to pronunciation difficulties, desired application features, and user preferences, which help guide the design and functionality of the proposed system فصيح.

Regarding pronunciation difficulties, 48.2% of participants reported confusion between similar letters such as ص (ṣād) and س (sīn), 33.1% reported difficulty in pronouncing full words correctly, and 30.9% indicated challenges with deep-throat sounds (such as ح، خ، ع، غ). These findings confirm that phoneme-level distinctions and articulation-specific sounds represent major areas of difficulty.

The questionnaire was also used to identify the most challenging letters for the children according to their parents, after listing all letters of the Arabic alphabet the results show that the most frequently reported problematic ones were ض(39.6%), ظ (33.1%) ,(%19.4) غ ,(%28.8) س ,خ (23.7%), ش and ص (18%). These results provide clear guidance on prioritizing specific phonemes within the system's training modules.

Concerning AI-based pronunciation evaluation, responses were distributed across the scale, with 32.4% selecting a neutral rating. However, when asked about the importance of immediate feedback after pronunciation, 63.3% rated it as highly important. This demonstrates a strong demand for real-time corrective feedback.

When asked whether the ability to track multiple children within one account is useful, 40.3% of participants answered "Yes," 46.8% selected "Maybe," and only 12.9% answered "No." Although not all participants strongly required this feature, the combined percentage of "Yes" and "Maybe" responses (87.1%) indicates a strong potential need for multi-child profile management. This suggests that supporting multiple child accounts under a single parent profile would add practical value, particularly for families with more than one child. From a technical perspective, this feature provides a higher feasibility, since children's accounts can be managed by a single parent account, which eliminates the need to create multiple ones for each child.

In terms of motivation mechanisms, 83.5% of participants believed that a reward-based system would encourage continued learning. Additionally, 56.1% rated progress tracking as highly important, and 64.7% stated that they would consider using an AI-based pronunciation application if it provided clear and accurate feedback. These findings highlight the importance of combining AI assessment with visible progress monitoring and motivational elements.

When asked about expected features, the most requested were immediate pronunciation correction (71.2%), progress tracking (64.7%), personalized training (52.5%), parental reports (40.3%), and a reward system (45.3%). These preferences indicate that users expect both technical accuracy and engaging design elements.

Based on these findings, the following key features will be implemented in the proposed system:

1. Immediate corrective feedback after each pronunciation attempt.
2. Individual progress tracking dashboard.
3. Support for multiple child profiles under one parent account.
4. Reward and motivation system to encourage continuous practice.
5. Targeted training exercises focusing on high-difficulty Arabic letters.

5.2.4 Observations

Structured observation sessions were conducted with 8 children using Alifbee as a benchmark application. The objective was to examine how children interact with a pronunciation-focused digital platform in a natural environment and to identify usability and engagement patterns.

Each child participated individually in a home setting, and a standardized evaluation checklist was used to ensure consistency. The observation focused on five main aspects: ease of use, interaction with the audio recording feature, response to feedback, overall engagement, and any technical challenges encountered. Attention was given to the child's ability to start activities independently, navigate the interface, understand when to record, interpret feedback, remain motivated, and respond to reward elements, while also documenting any confusion or system delays.

These observations provided practical insights into children's real-time behavior, supporting the development of a more intuitive and engaging learning experience in our proposed system.

5.2.5 Demographics of participants

This table presents the demographic information of the participants involved in the observations, including their name, age, and gender. All participants were native Arabic speakers, and the application interface and instructional content were presented in Arabic. None of the participants had any known learning or developmental conditions (e.g., ADHD or dyslexia) that could influence their pronunciation performance or system interaction.

Name	Age	Gender
Atheer	6	Female
Sahab	7	Female
Adeem	7	Female
Fahad	9	Male
Mohammed	10	Male
Aleen	11	Female
Alma	12	Female

Abdullah	13	Male

Table 7: Demographics of participants (Observation)

5.2.6 Observation results

The conducted observations provided several valuable insights. The following section summarizes our findings based on five significant points, describing the children's experience during their interaction with AlifBee App. All observation details and questions are provided in Appendix B.

Ease of use

Overall, the pronunciation assessment application demonstrated a generally high level of ease of use, particularly among children aged 9–13. Most participants in this age group were able to independently start the pronunciation exercise, record their voice without prior explanation, and navigate between screens with minimal or no assistance. The microphone icon was widely recognized as intuitive, and the bottom navigation bar supported smooth movement between pages for most older children.

However, some usability challenges were observed. A few children (particularly Fahad and Abdullah) required initial exploration time before identifying how to start the exercise. Similarly, Aleen and Adeem experienced brief confusion regarding where to begin, indicating that the entry point to the first exercise could be made more visually prominent.

Younger participants, especially Atheer (6 years old), required significant guidance to initiate the exercise, record audio, and navigate between screens. Adeem (7 years old) also faced difficulty understanding when to stop the recording and occasionally interacted randomly with the interface. These findings suggest that children under 8 may benefit from clearer visual cues, simplified interface options, or guided onboarding support.

Audio Recording Interaction

The observations indicate that the audio recording feature was generally understandable and easy to use for most children, particularly those aged 9–13. Older participants clearly recognized that pressing the microphone button will initiate the recording and understood that results would appear after completing their pronunciation. Most of them demonstrated confidence, showed little to no hesitation before recording, and followed the recording flow correctly.

However, minor challenges were observed among younger children (ages 6–7). Atheer required demonstration to understand when to start and stop recording, and Adeem occasionally stopped the recording too early or clicked randomly before results were displayed. Mohammed also showed slight difficulty in understanding how to properly stop the recording, even though he understood how to start it. These findings suggest that while the recording initiation is intuitive, the stopping mechanism may require clearer visual or auditory cues.

Interaction with Feedback

The observations indicate that most children paid attention to the corrective feedback provided after each pronunciation attempt. Older participants (ages 9–13) were generally able to understand whether their answer was correct or incorrect and, in several cases, used the feedback to improve subsequent attempts. For example, Alma immediately repeated mispronounced words to correct them, and Abdullah applied feedback to avoid repeating mistakes. This suggests that the immediate result display (correct/incorrect with visual and sound cues) is effective.

However, the depth of understanding of detailed feedback varied. Some children (e.g., Fahad and Mohammed) needed a clearer indication of which specific letter or sound was mispronounced. In these cases, feedback that showed only the correct answer, without comparison to the child’s response, was not always sufficient. Younger participants, especially Atheer (6 years old), struggled to interpret written feedback and relied primarily on visual indicators such as colors, symbols, or animations.

Another recurring observation was that several children did not pay attention to the final overall feedback page (summary of performance). While they understood question-by-question results, they often skipped or overlooked the overall performance breakdown unless prompted. This suggests that cumulative feedback may need to be more visually engaging or simplified.

Engagement level

The observations show that the application successfully captured and maintained engagement for most children, particularly during the initial stages of the assessment. The majority of participants demonstrated enthusiasm, focus, and willingness to complete the exercises. Older children (ages 9–12) generally maintained attention throughout the session and completed tasks sequentially without distraction.

However, slight declines in engagement were observed in some cases. Fahad and Abdullah showed reduced attention when tasks became repetitive, and Mohammed and Abdullah began to lose enthusiasm toward the end of the session. Similarly, Atheer (6 years old) displayed initial excitement, but lost focus more quickly compared to older participants. These findings suggest that task variety and session length may influence sustained engagement.

Regarding curiosity and interaction, some children asked clarification questions, particularly when feedback was unclear (e.g., Fahad, Mohammed, Adeem). Others completed the exercises quietly without asking questions, indicating confidence or comfort with the activity.

The streak and reward system produced mixed responses. Some children (e.g., Sahab, Fahad, Abdullah, and Atheer) responded positively to rewards and appeared motivated by points, stars, or achievements. In contrast, older participants, such as Alma, Aleen, and Adeem, showed little interest in the reward system and paid little attention to earned points. This suggests that gamification elements may be more motivating for younger users.

Technical Issues

Overall, the application functioned smoothly for most participants, with minimal technical disruptions during the assessment process. Most children did not experience noticeable delays while navigating between screens or receiving results. This indicates stable system performance in typical home environments.

However, minor issues were observed in specific cases. Abdullah experienced brief delays during transitions between questions, and the speaker icon (audio playback) showed slight lag before playing the word. Additionally, some children (such as Fahad and Atheer) appeared confused during short waiting moments before results were displayed, even when no actual technical delay occurred. This suggests that perceived delay may sometimes be related to a lack of clear loading indicators rather than system performance.

Understanding of system messages was another area of concern, particularly among younger children. Atheer struggled with text-based notifications and relied more on visual cues (colors, symbols, animations). Adeem also faced difficulty understanding why certain exercises were locked or inaccessible.

5.3 User Interactions

This section presents the Use Case Diagram of the Faseh (فصيح) application, illustrating the main interactions between the users and the system functionalities.

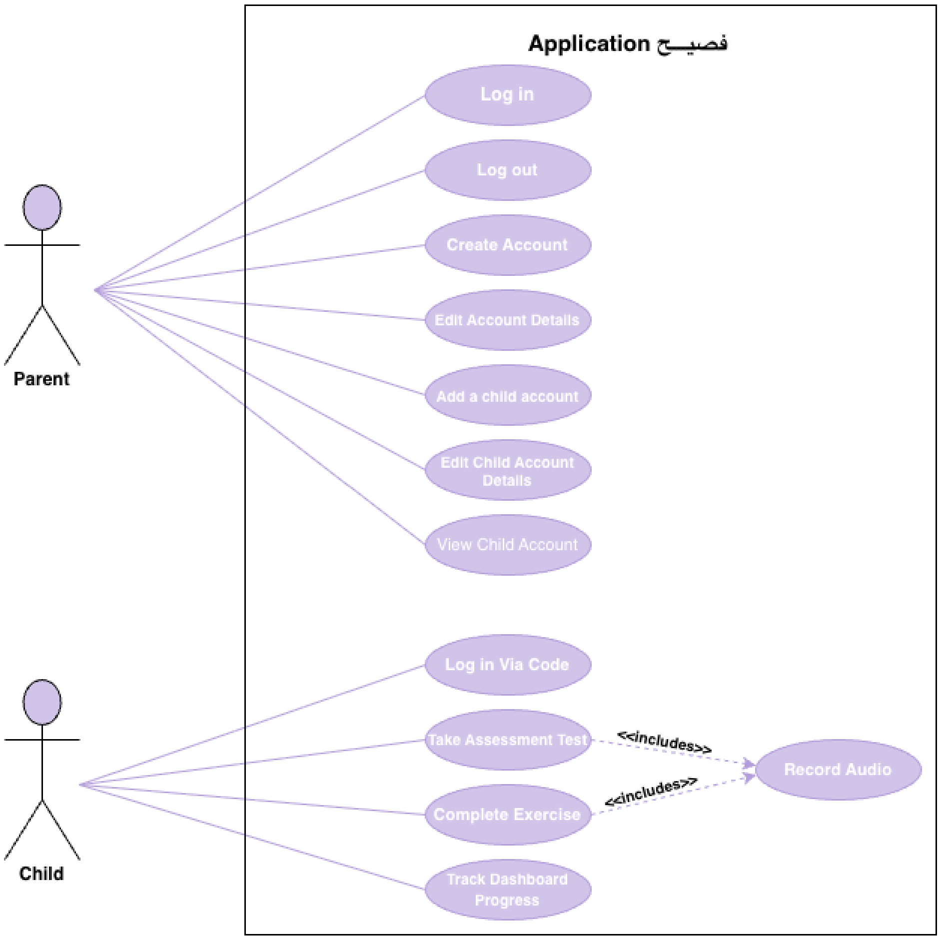


Figure 27: The system's Use Case Diagram

5.4 Product Backlog

This section presents the Product Backlog of the proposed system, which outlines a prioritized list of functional and non-functional requirements required for system development across the planned releases.

Example Definition of Ready	
✓	Business value is clearly articulated
✓	Details are sufficiently understood
✓	Dependencies are identified; no blocking dependencies exist
✓	Team is appropriately staffed relative to the PBI
✓	Estimated and small enough to be completed during sprint
✓	Acceptance criteria are clear and testable
✓	Performance criteria, if any, are defined and testable
✓	Team understands how to demo the completed PBI

Product Backlog Table

In this section we present our product backlog, which is organized in the form of user stories that describe system features from the users' perspective, along with their corresponding size, type, and acceptance criteria. These backlog items (PBIs) represent all anticipated work needed to deliver value to end users, including new features, usability improvements, and technical enhancements.

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work,	Status (To do, in progress, or done)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.

			knowledge acquisition)		
1	As a parent, I want to create an account so that I can manage my child's learning.	3	Feature	Done	• As a parent, if I enter a valid email, password and full name, then my account should be created successfully. • As a parent, if registration is successful, then I should receive a success message indicating that the account has been created successfully and be redirected to the Create Child Profile page. • As a parent, if I enter an email that is already registered, then an error message should be displayed. • As a parent, if I enter an invalid email format, then an error message should be displayed. • As a parent, if I leave the email or password field empty, then an error

					message should be displayed. • As a parent, if the password does not meet the required security criteria (minimum 8 characters, including letters, numbers, and unique symbol), then the system should display a validation message and prevent account creation.
2	As a parent, I want to log in so that I can access my account settings.	2	Feature	Done	• As a parent, if I enter the correct email and password, then I should be logged in successfully and redirected to the parent dashboard. • As a parent, if I enter a correct email but incorrect password, then an error message should be displayed. • As a parent, if I enter an unregistered email, then an error message should be displayed. • As a parent, if I leave the email or password field

					empty, then an error message should be displayed. • As a parent, if I forget my password, then I should be able to reset it.
3	As a parent, I want to be able to log out from my account so that I can ensure the security of my account and my child's information.	2	Feature	Done	• As a logged-in parent, if I click on the Logout button within the application, then I should be successfully logged out from my account. • If I reopen the application after logging out, then I should be required to enter my credentials again to access my account.
4	As a parent, I want to edit my profile information so that I can keep my account details accurate and up to date.	3	Feature	Done	• As a logged-in parent, if I access the profile settings page and update my information (e.g., name, password), then the system should successfully save the changes. • If I successfully update my profile, then I should receive a confirmation

					message indicating that the changes were saved.
5	As a parent, I want to create a child profile so that my child can use the application.	3	Feature	Done	• As a parent, the system should allow me to create multiple child profiles under my account. • As a parent, if I add a new child profile, then the system should require the child's name, dob, gender and avatar. • As a parent, if a child profile is created successfully, then it should be linked to my account. • As a parent, if required fields are missing, then the child profile should not be created, and an error message should be shown.
6	As a parent, I want to manage my child's profile so that I can monitor their learning progress.	5	Feature	In progress	• As a parent, if I have multiple child profiles, then I should be able to view each profile separately.

					• As a parent, if I edit the child's information, then the updated data should be saved successfully. • As a parent, if I choose to delete the child profile, then the system should remove it after confirmation. • As a parent, if I switch between child profiles, then the displayed data should update accordingly.
7	As a child, I want to record my voice reading an Arabic Words so that the system can assess my pronunciation.	5	Feature	In progress	• As a child, if I click the record button, then the system should start recording my voice. • As a child, if I click the stop button, then the system should stop recording. • As a child, if my recording is successful, then the system should confirm that my voice has been recorded. • As a child, if my microphone is not enabled, then the system

					should display an error message.
8	As a child, I want to receive feedback on my pronunciation accuracy so that I know which letters I mispronounced.	8	Feature	In progress	• As a child, if I submit a voice recording, the system should automatically analyse my pronunciation. • As a child, if the pronunciation analysis is completed, then I should receive immediate feedback including my score along with my strong and weak letters. • As a child, if I mispronounce a letter, then the system should provide suitable feedback. • As a child, if I pronounce a letter correctly, then the system should indicate that it is correct in the feedback. • As a child, I want the feedback to be simple and easy to understand.
9	As a child, I want to complete exercises so	6	Feature	In progress	• As a child, when I open exercises, I should see

	that I can improve my pronunciation.				only the letters I need to practice. • As a child, when I select a letter, I should see its exercise levels. • As a child, when I start a level, I should get questions that match my level for that letter. • As a child, I should complete the current level before unlocking the next level. • As a child, when I finish a level, I should see my grade. • As a child, when I complete all levels, the letter should be marked as completed. • As a child, incomplete levels should not be marked as completed.
10	As a child, I want to track my progress so that I can see my improvement over time.	5	Feature	In progress	• As a child, if I have not completed any exercises, then the system should display that no progress is available yet. • As a child, if I take a Reassessment Test and my level changes, then

					the system should update my level displayed in the progress section.
11	As a child, I want to earn points so that I stay motivated to practice.	3	Feature	In progress	• As a child, if I complete an exercise activity, then the system should add points to my account. • As a child, if points are awarded, then the system should display the updated points. • As a child, if I earn points, then the system should update my ranking on the leaderboard and allow me to view my position.
12	As a child, I want my daily progress to be maintained through a streak so that I stay motivated to continue learning regularly.	3	Feature	In progress	• As a child, if I complete at least one exercise in a day, then the system should count that day toward my streak. • As a child, if I practice on consecutive days, then the system should increase my streak count. • As a child, if I miss a day without completing any

					exercise, then the system should reset my streak. • As a child, if my streak is updated, then the system should display the current streak clearly on my home screen. • As a child, if my streak increases, then the system should update my ranking on the leaderboard and allow me to view my position.
13	As a Child, I want to enter a simple 6-digit code provided by my parent, so that I can access my personalized learning profile on my own device without needing a username or password.	3	Feature	Done	• As a child, if the entered code is correct and hasn't expired, then the system should link my device to my specific profile and navigate me directly to my home screen. • As a child, if I enter an incorrect or expired code, then the system should display a friendly error message in Arabic and allow me to try entering a new code.
Non-Functional Requirements					

14	**Security** As a parent, I want to log into my account securely using valid information, so that I can access my personal data without unauthorized access.	4	Feature	Done	• If a parent enters incorrect login credentials (wrong password or email), then the app displays an error message and does not grant access.
15	**Performance** As a child, I want my pronunciation feedback to be generated within 15 seconds of submitting my recording so that I can receive immediate correction.	8	Feature	In progress	• If a child submits a recording, then the feedback is displayed within 15 seconds. • If the backend processing exceeds 5 seconds, then the app shows a friendly loading animation with a message.
16	**Reliability** As a child, I want the application to function reliably without crashing, ensuring a smooth and uninterrupted learning experience.	8	Feature	Done	• If a child uses the app continuously for 10 minutes, then the app should not crash more than once during that session. • If tested across 100 practice sessions, then the crash rate should be below 2%. • If a child completes a pronunciation exercise,

					then their progress is automatically saved before moving to the next activity.
17	**Availability** As a user, I want the app to be accessible 95% of the time, so that I can practice whenever I want without interruption.	8	Feature	Done	• If monitored over a period of time, then the app's uptime should be at least 95% (allowing for up to 36 hours of downtime total). • If the app is unavailable due to maintenance or server issues, then users see a clear message explaining that.
18	**Usability** As a child, I want the app interface to be simple and easy to navigate within 5 minutes of use so that I can focus on learning Arabic.	8	Feature	Done	• If a first-time child user opens the app, then they should be able to complete their first recording attempt within 3 minutes without assistance. • If tested with 10 children (ages 5–13), then at least 8 of them should be able to navigate to the pronunciation practice screen without help. • If surveyed after first use, then at least 85% of

					children should rate the app as "easy to use".
Enabler Stories					
19	As an ML engineer, I need to preprocess the user's recorded audio so that the pronunciation model can analyze consistent and noise-reduced input.	8	Technical Work	In progress	• If an audio file is uploaded, then silence trimming is applied. • If background noise exceeds a threshold, then noise reduction is applied. • If preprocessing fails, then the system logs an error. • If the processed audio duration is less than 0.2 seconds, then the system should return the original audio file to avoid losing valid speech.
20	As an ML engineer, I want to integrate a pre-trained pronunciation model and fine-tune it using a children's speech dataset so that the system can better understand children's voices.	8	Technical Work	In progress	• If a pre-trained Arabic pronunciation model is selected, then it should be successfully integrated into the system pipeline. • If the dataset is ready, then it should be converted into a suitable format for model fine-tuning.

					• If fine-tuning is performed, then the model should be trained using children's Arabic speech data. • If fine-tuning is completed, then the final fine-tuned model and processor should be saved for later use in the assessment pipeline.
21	As an ML engineer, I need to train a pronunciation assessment model so that the system can detect mispronunciations.	13	Technical Work	In progress	• If training data is provided, then the model trains without crashing. • If evaluated on validation data, then accuracy meets predefined benchmark. • If the model analyzes a recording, then it should return the results in a clear and organized format.
22	As a developer, I need to deploy the trained pronunciation model as an API service so that the app can request feedback.	5	Technical Work	In progress	• If a recording is submitted, then the API receives the audio file. • If inference is completed, then pronunciation results are returned in JSON format.

					• If API fails, then an error is logged and a fallback message is returned.

Table 8: The system's product backlog

6 System Design

6.1 Architectural Diagram

The architecture of the Faseh (فصيح) application is designed to provide an effective and structured pronunciation learning experience for Arabic language learners. After evaluating different architectural approaches, the system adopts a client–Server architecture combined with a Layered design. The Client–Server model separates the user-facing mobile application from the backend services responsible for processing and data management, while the layered design organizes system responsibilities into distinct logical layers.

The client side focuses on user interaction, audio recording, and feedback presentation. On the server side, system functionality is divided into layers, including an application layer for request handling and orchestration, an AI processing layer for pronunciation analysis and scoring, and a data layer for storage and retrieval of user information and audio files. This architectural combination enhances system scalability, maintainability, and flexibility, while enabling future improvements to the pronunciation assessment models without affecting the client interface.

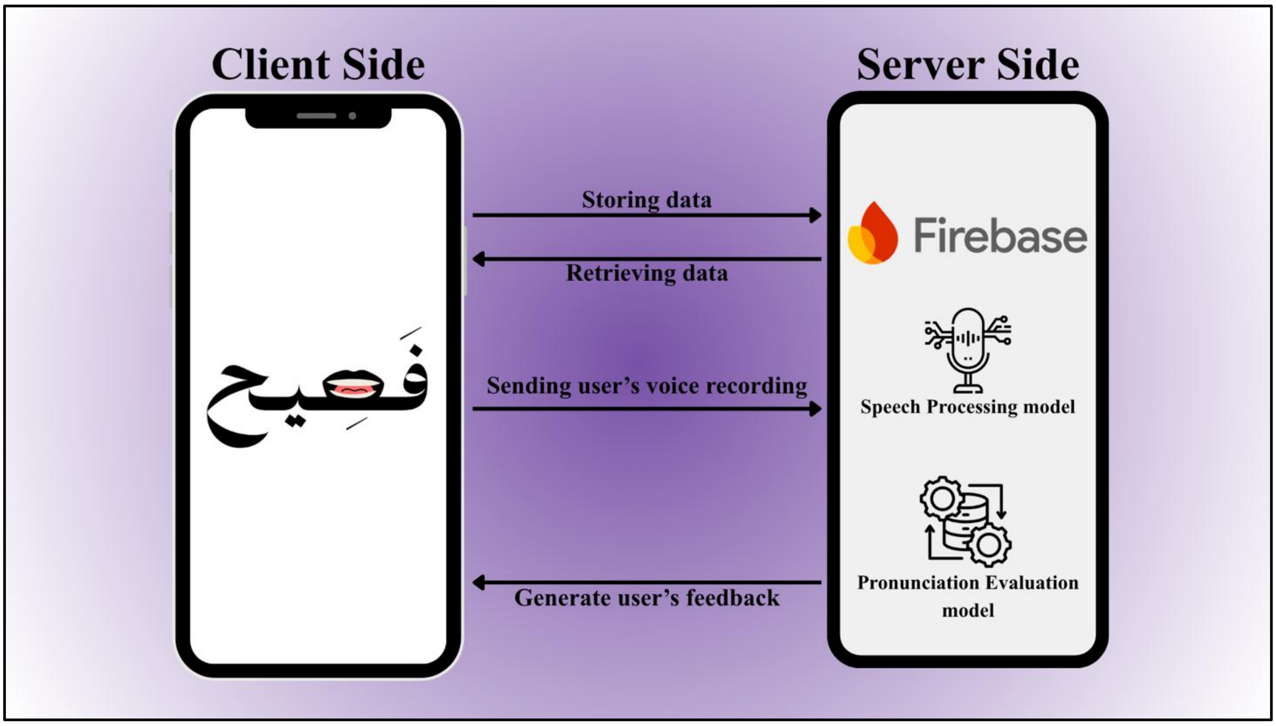


Figure 28 :Faseh (فصيح) Architecture

System Components

The system is divided into two main components:

- **Client Side**

The client side represents the user-facing part of the application and is responsible for the following:

- **Pronunciation Practice Interface:**

  Allows learners to read the displayed Words, record their pronunciation, and submit audio recordings for evaluation.

- **Feedback Visualization:**

  Displays pronunciation scores, highlighted pronunciation issues, and suggested exercises in a clear and understandable format.

- **Progress Dashboard:**

Enables learners to view their pronunciation history, and completion status of exercises.

- **Intuitive User Interface:**

Ensures a simple and intuitive experience that allows learners to focus on pronunciation practice without technical complexity.

- **Server Side**

The server side manages the core logic and data operations of the system. It ensures smooth execution of the following key features:

- **User Authentication and Account Management**

Handle user registration and login using Firebase Authentication, allowing learners to access their personal progress and saved results.

- **Database and Cloud Storage:**

Stores user profiles, scores, feedback results, and learning progress, as well as audio recordings securely.

- **Speech Processing Module:**

Processes uploaded audio recordings to prepare them for pronunciation analysis.

- **Pronunciation Evaluation Module:**

Applies a developed pronunciation scoring model to analyze learner's pronunciation and generate pronunciation scores.

- **Feedback Generation Module:**

Converts pronunciation evaluation results into meaningful feedback and recommends suitable exercises based on learner performance.

- **Data Management Services:**

Handles all requests from the client, including saving evaluation results, retrieving progress data, and managing user sessions.

Input and Output Flows

Inputs

- **Voice Input:**

  Audio recordings captured from the learner to evaluate Arabic pronunciation accuracy.

- **User Interaction Input:**

  Actions such viewing feedback, view user's profile and accessing progress information.

Outputs

- **Textual and Visual Feedback:**

  Numerical evaluation and graphical representation of the learner's pronunciation performance.

- **Progress Reports:**

  Displays learner improvement over time through stored performance data.

Users and External Systems

Users

- **Children:**

  The primary users who interact directly with the Faseh (فصيح) application to practice and improve their Arabic pronunciation.

- **Parents:**

  The secondary users who manage the registration process, supervise their child's learning journey, and monitor pronunciation progress.

External Systems

- **Cloud Storage and Database Services:**

  Used for secure storage and retrieval of audio files and learner data.

- **Speech Recognition and Processing Models:**

  Pre-trained models used to analyze pronunciation and support automated evaluation.

6.2 Class Diagram /DFD

In this section, we present the class diagram for the Faseh (فصيح) application. The class diagram provides an object-oriented view of the system by showing the main classes, their attributes, methods, and relationships. It helps explain how the application components interact with each other and how the system logic is organized.

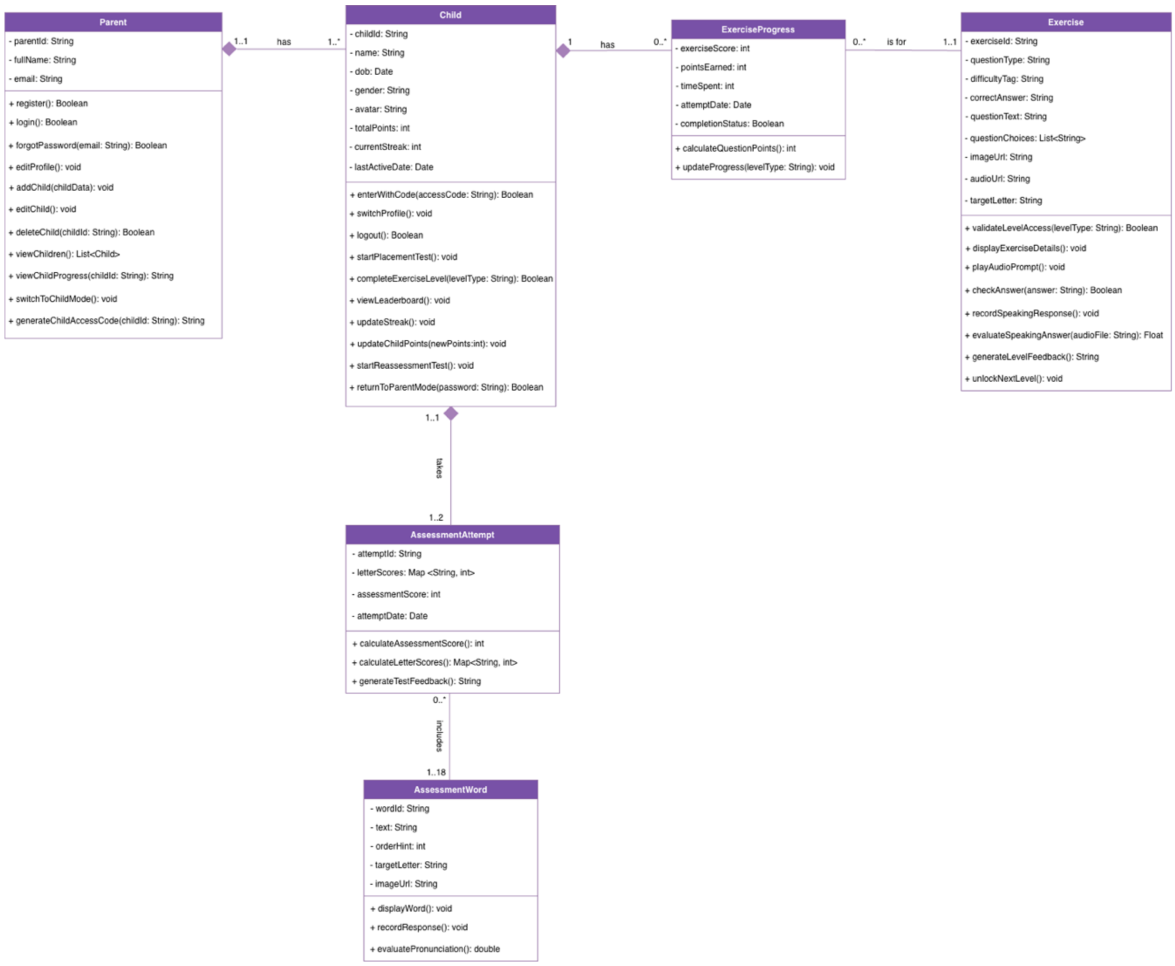


Figure 29: Class Diagram

Parent and Child Classes

The Parent class represents the parent user who is responsible for registering, logging in, and managing child profiles. The methods in this class reflect the actions that the parent can perform, such as managing account information, creating child profiles, viewing children, checking child progress, and switching to child mode.

The Parent class is connected to the Child class through a composition relationship, which shows that the child profile is strongly linked to the parent account. Each parent has one or more children, while each child belongs to exactly one parent. This supports the application's registration logic, where a parent must create a child profile to complete the registration process.

The Child class represents the child user who interacts with the learning side of the application. This class supports actions such as entering the application using an access code, starting the placement test, completing exercise levels, viewing the leaderboard, updating points and streaks, and starting a reassessment. Since most learning activities are performed by the child, this class is central to the application logic.

AssessmentAttempt and AssessmentWord Classes

The AssessmentAttempt class represents an assessment taken by the child. It is used to store and manage the result of each assessment attempt, including the child's performance in the target letters and the overall assessment result. The class also includes behavior related to calculating scores and generating feedback.

The Child class is connected to the AssessmentAttempt class through a composition relationship, which means that assessment attempts belong to a specific child. A child can have one to two assessment attempts, representing the initial placement assessment and the reassessment.

The AssessmentWord class represents the predefined words used during the pronunciation assessment. These words are used to display assessment content, record the child's response, and evaluate pronunciation. The AssessmentAttempt class is connected to the AssessmentWord class through the "includes" relationship, meaning each assessment attempt includes a set of assessment words, while the same assessment words can be reused across multiple attempts by different children.

Exercise and ExerciseProgress Classes

The Exercise class represents the learning exercises provided to the child after the assessment. It supports different exercise types, such as MCQ, listening, and speaking exercises, and connects each exercise to a specific target letter and difficulty level. The methods in this class represent the main exercise actions, such as displaying exercise content, playing audio prompts, checking answers, recording speaking responses, evaluating pronunciation, generating feedback, and unlocking the next level.

The ExerciseProgress class represents the child's progress in a specific exercise. It stores and manages the progress information needed to determine whether an exercise has been completed, how many points were earned, and when the attempt occurred. This class also supports updating progress after the child completes an exercise.

The Child class is connected to the ExerciseProgress class through a composition relationship because the progress record is created for a specific child. A child may have many progress records, with each record representing the child's progress in one exercise. The ExerciseProgress class is connected to the Exercise class through the "is for" relationship, which means that each progress record is linked to one specific exercise. However, the same exercise can appear in many progress records because different children may complete the same exercise. This design allows the system to save completion information for each exercise separately, then use these records to check whether the child has completed all exercises required for a level or section.

6.3 Component Level Design

6.3.1 Add Child Flow

The flowchart illustrates the process of adding a child after the parent registers in the application. The process begins with the parent completing the registration process. Once registered, the system automatically directs the user to the "Add Child" feature.

The user is then required to enter the child's information, including the name, gender selection, date of birth, and avatar selection. After entering the details, the system checks whether the inputs are correct.

If the inputs are incorrect, the user is redirected to re-enter the information. If the inputs are valid, the child is successfully added to the system.

Afterward, the user is given the option to add more children. If the user chooses to add another child, the process repeats. Otherwise, the process ends.

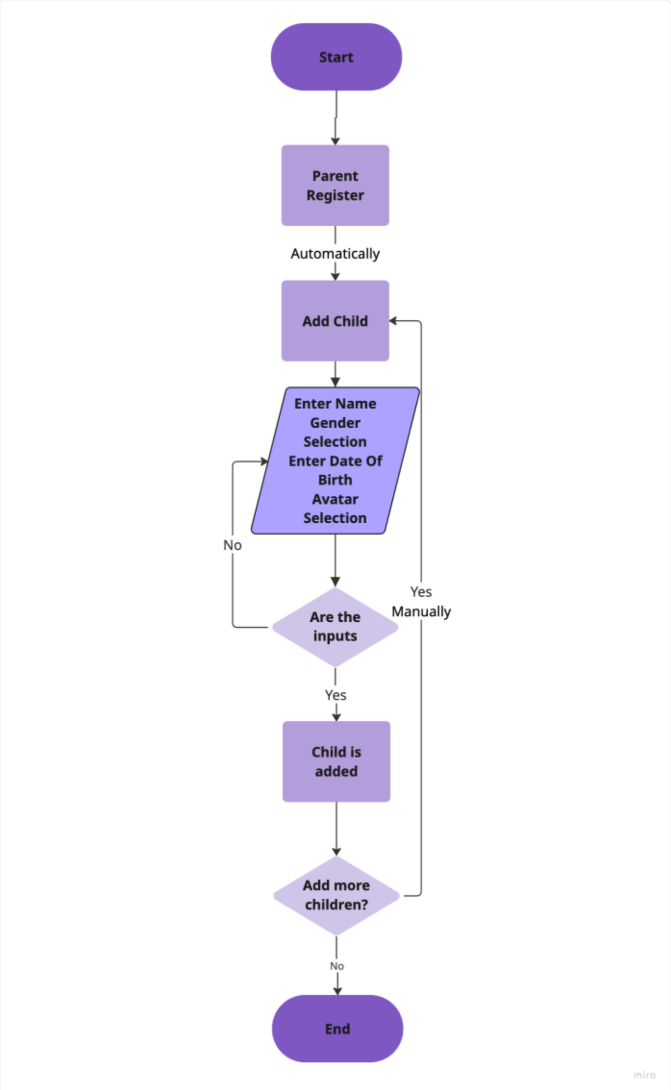


Figure 30:Add Child (Flowchart)

6.3.2 Placement Test Activity diagram

The activity diagram illustrates the process of conducting the placement test which evaluates the child's ability to pronounce each letter in different positions within a word (beginning, middle, and end). The process begins by displaying a test item for the child to pronounce while the system stores the audio data. Then the audio is sent to the model to analyze pronunciation and calculate a score, while audio features are extracted and results are saved.

The system then checks whether there are more test items. If additional items remain, the process repeats until all items are completed. Once the test is finished, the system generates the placement result, identifies weak letters, and recommends personalized exercises for the child.

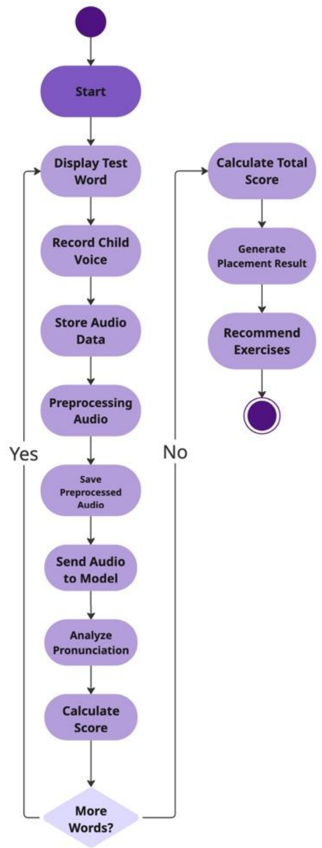


Figure 31: Placement Test (Activity diagram)

6.3.3 Exercise Completion Flow

The flowchart illustrates the process of a child completing an educational exercise within the application. The process begins with the child entering the exercise section and selecting a specific letter to practice. Once a letter is chosen, the child progresses through three sequential levels: MCQ, Listening, and Speaking.

For each level, the system evaluates the child's performance against a minimum score threshold of 60%. If the child scores below 60%, they are redirected to repeat the same level using the same set of questions. This allows the child to focus on correcting the specific mistakes made in that level and ensures that the improvement is based on learning and retrying the same content rather than being exposed to a different set of questions. If the child achieves a score of 60% or higher, the level is considered successfully completed, and the next level is unlocked.

After completing all three levels, the system updates the child's progress in the database and marks the selected letter as mastered. The process concludes once the progress is saved, and the child is returned to the main dashboard.

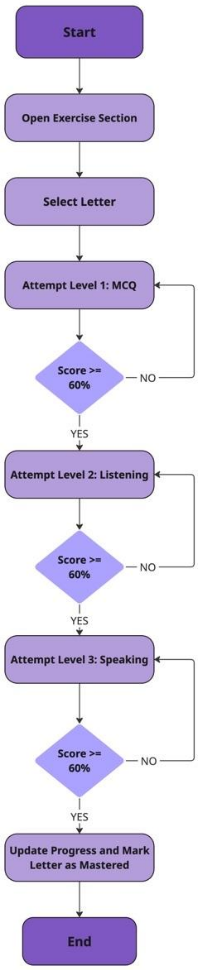


Figure 32: Complete Exercise (Flowchart)

6.4 Data Design

6.4.1 Data Models

6.4.1.1 *The ER Diagram*

In this section, we present the Entity Relationship (ER) diagram for the Faseh (فصيح) application, which illustrates the main data entities, their attributes, and the relationships between them. An Entity Relationship (ER) diagram is a conceptual data modeling tool used to represent the main entities in a system, their attributes, and the relationships between them. It helps clarify how data is organized and how different parts of the system interact from a database perspective.

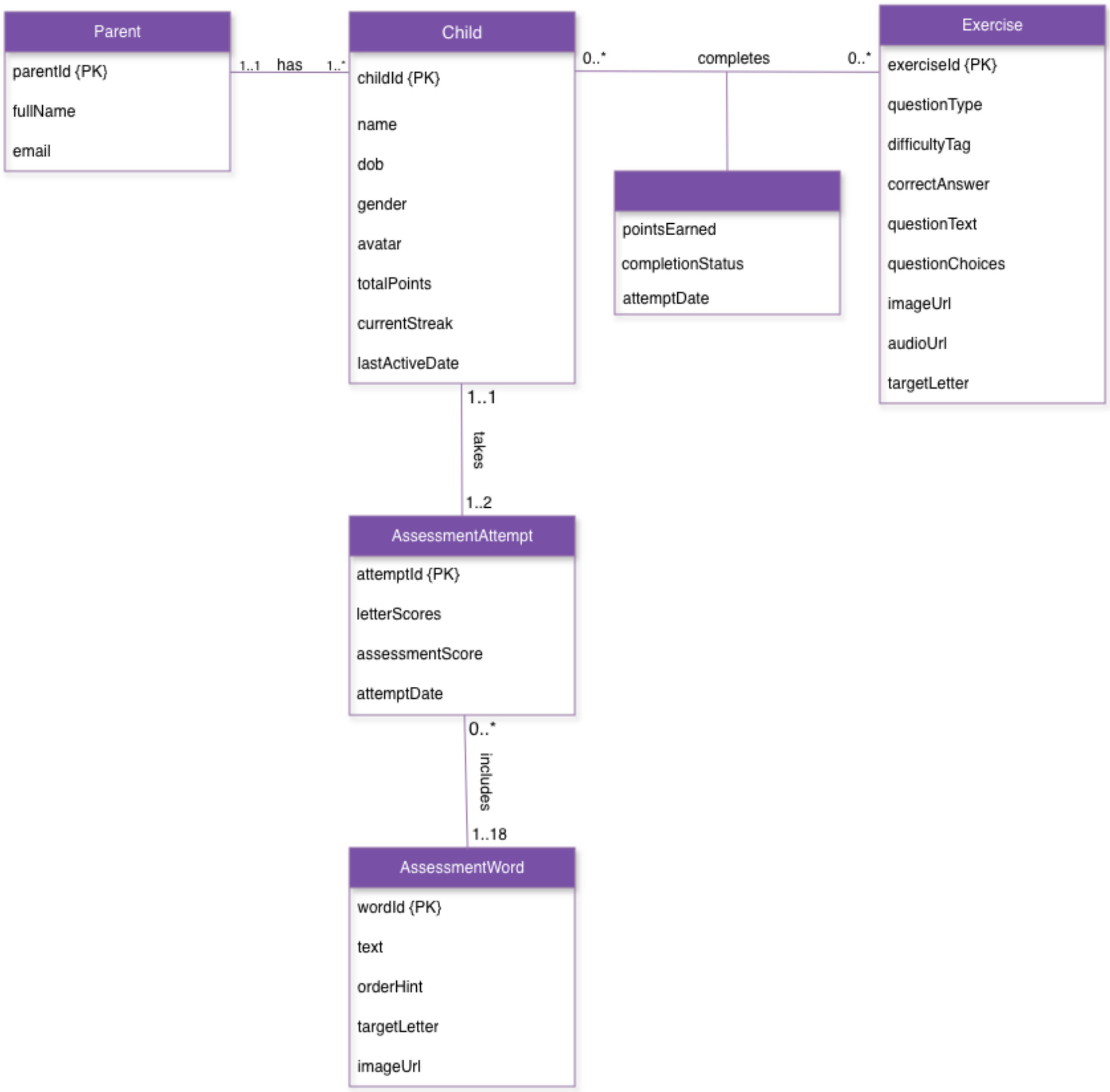


Figure 33: ER Diagram

The ER diagram represents the main data structure of the Faseh (فصيح) application and shows how the system stores and manages information related to parents, children, assessments, exercises, and progress tracking.

Parent Entity

The Parent entity represents the parent user who registers and manages child profiles in the application. Each parent has attributes such as parentId, fullName, and email. A parent is connected to the Child entity through the “has” relationship. The relationship shows that each parent must have one or more children, while each child belongs to exactly one parent. This reflects the application’s registration flow, where the parent must create a child profile to complete the registration process.

Child Entity

The Child entity represents the child user who uses the application to complete pronunciation assessments and exercises. It includes attributes such as childId, name, dateOfBirth, gender, avatar, totalPoints, currentStreak, and lastActiveDate. The child is the central entity in the system because most learning activities, progress records, and assessment results are linked to the child profile. The totalPoints attribute is used to support the leaderboard and encourage healthy competition between children, while currentStreak tracks the number of continuous days the child practices, helping promote consistent learning and engagement.

Assessment Attempts and Assessment Words Entities

The child is connected to the AssessmentAttempt entity through the “takes” relationship. This relationship indicates that each child can take one to two assessment attempts. The first attempt represents the initial placement assessment, while the second can represent the reassessment used to measure improvement. The AssessmentAttempt entity stores the results of each assessment attempt using attributes such as attemptId, letterScores, assessmentScore, and attemptDate. This structure helps the system show the child’s progress by keeping each assessment attempt with its related scores and date, making it easier to compare performance between the initial assessment and the reassessment.

The AssessmentAttempt entity is connected to the AssessmentWord entity through the “includes” relationship. Each assessment attempt includes a set of 18 assessment words. This number was selected because the assessment targets the six Arabic letters supported by the app, each letter is included in three words. These three words place the target letter in different positions: at the beginning, in the middle, and at the end of the word. This design allows the system to evaluate the child's pronunciation of each target letter in multiple word positions, making the assessment more comprehensive and fair.

The AssessmentWord entity represents the predefined words used during the pronunciation assessment. It includes attributes such as wordId, text, orderHint, targetLetter, and imageUrl. These words are used to evaluate the child's pronunciation of the selected target letters.

Exercise and Progress Tracking

The Exercise entity represents the exercises provided to the child after the assessment. It contains attributes such as exerciseId, questionType, difficultyTag, correctAnswer, questionText, questionChoices, imageUrl, audioUrl, and targetLetter. These attributes allow the system to store different types of exercises, such as MCQ, listening, and speaking exercises, while also linking each exercise to a specific difficulty level and target letter.

The relationship between Child and Exercise is represented by the "completes" relationship. Since one child can complete many exercises, and one exercise can be completed by many children, this is a many-to-many relationship. The relationship includes progress-related attributes such as pointsEarned, completionStatus, and attemptDate. These attributes allow the system to track whether a child completed a specific exercise, when it was completed, and how many points were earned. This structure supports the application's progress tracking logic, where each exercise completion is recorded separately and later used to determine whether a level or exercise section has been completed.

Overall, the ER diagram provides a clear database-level view of the application by showing how parent, child, assessment, exercise, and progress data are connected. It supports the main flow of the application, from creating a child profile to recording assessments, completing exercises, and tracking progress.

6.4.1.2 *The Non-Relational Data Model*

Our application Faseh (فصيح), uses the NoSQL database due to its primary requirements such as real-time UI updates, rapid database retrievals for gamified learning, and a flexible hierarchy for parent-child profiles, therefore the NoSQL document database was the optimal choice for our application. The non-relational data model below illustrates how data is organized within our structure, with entities and objects are represented by collections and documents, allowing for smooth expansion and adaptation.

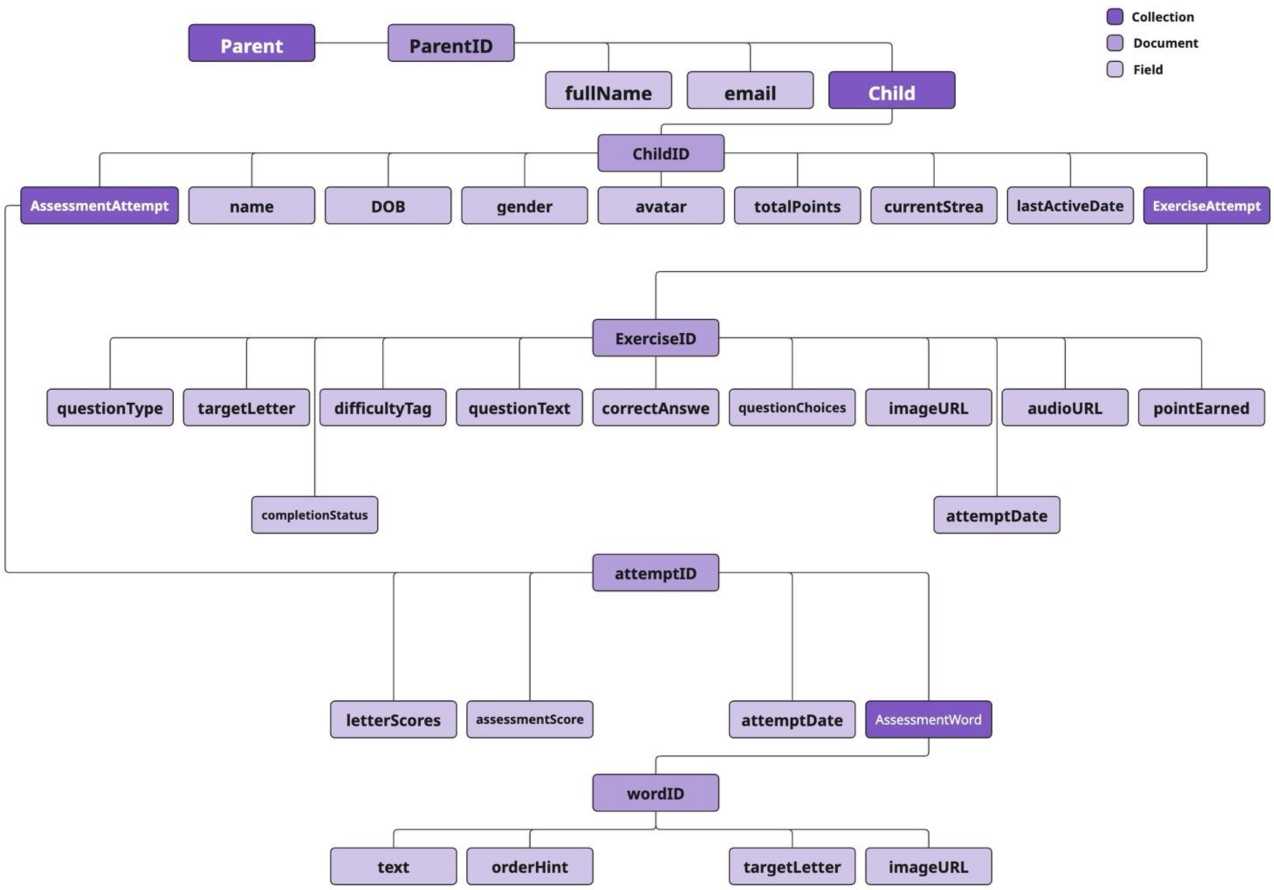


Figure 34: The Non-Relational Data Model

6.4.2 Data Collection and Preparation

This part is not applicable in our application.

6.5 Interface Design

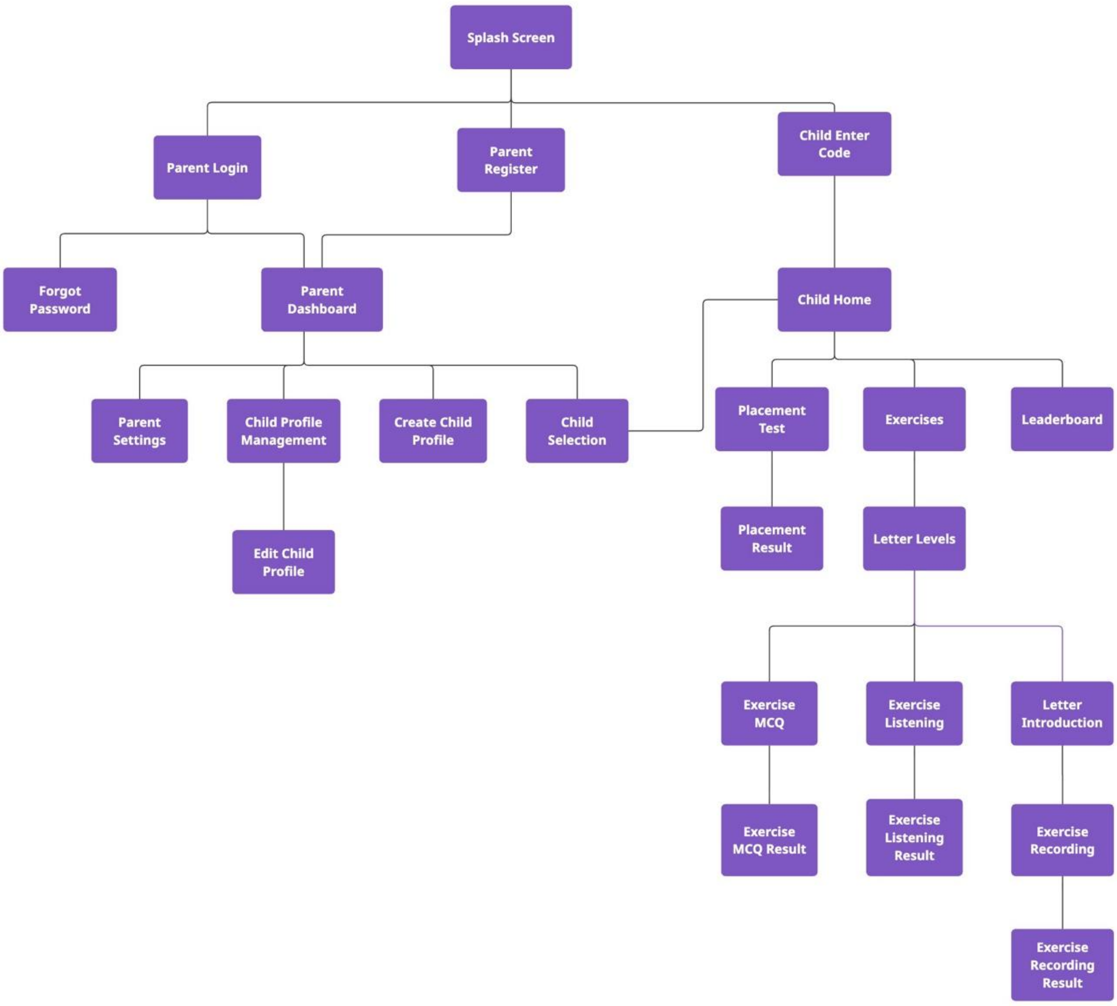


Figure 35:Navigation Diagram

This navigation diagram illustrates the core structure of Faseh ((فصيح application, designed to support Arabic pronunciation learning for children. It outlines how users navigate through key sections such as parent login and registration, child access via code entry, child profile management, and structured exercises.

The interface was carefully designed to ensure smooth and intuitive navigation for both children and parents, while maintaining a clear separation between user roles and supporting a guided learning experience.

The following UX guidelines were incorporated during the interface design process:

1. Simplicity and Clarity

The interface prioritizes simplicity by using clean layouts, minimal text, and clearly defined actions. This aligns with Nielsen's heuristic of *"aesthetic and minimalist design"[23]*, which emphasizes reducing cognitive load by eliminating unnecessary elements. The design ensures that children can easily understand what to do without confusion, especially during exercises and navigation between screens.

2. Consistency and Standards

A consistent visual language was maintained across the application, including colors, button styles, and layout structures. For example, the use of a unified purple color scheme and consistent button shapes helps users quickly learn interaction patterns. This follows Nielsen's principle of *consistency and standards[23]*, allowing users to predict system behavior.

3. Clear Navigation Flow

The application is structured with a clear separation between parent and child flows. Parents manage profiles and settings, while children interact with exercises and learning content. This logical structure ensures intuitive navigation and reduces user confusion, especially when switching between different roles.

4. Immediate Feedback

The system provides instant feedback after completing activities such as MCQ, listening, and pronunciation exercises. This reflects Norman's principle of *feedback*, where users are informed about the results of their actions. Immediate feedback is essential for learning, as it helps children recognize mistakes and improve their pronunciation.

5. Child-Friendly Design

The interface is designed specifically for children by using large buttons, simple language, and visually engaging elements. This aligns with user-centered design principles, ensuring that the interface matches the cognitive and motor abilities of young users, making interaction easier and more enjoyable.

6. Error Prevention and Confirmation

The system incorporates confirmation mechanisms such as pop-up messages (e.g., confirming logout actions or other critical operations) to prevent unintended actions. These mechanisms allow users to review their decisions before proceeding, reducing the likelihood of errors

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8 Appendix

8.1 Appendix A: Questionnaire Questions

كم عمر طفلك؟ *

4–6 سنوات

7–8 سنوات

9–12 سنة

13–15 سنة

كم عدد الأطفال لديك؟ *

طفل واحد

طفلان

3–4 أطفال

5 أطفال فأكثر

Figure 36: Questionnaire Q1 and Q2

ما هي اللغة التي يتحدث بها طفلك في المنزل؟ *

العربية

الإنجليزية

يتحدث بالعربية والإنجليزية، لكن غالبًا العربية

يتحدث بالعربية والإنجليزية، لكن غالبًا الإنجليزية

هل سبق وأن استخدمت تطبيقات لتعليم طفلك الأحرف العربية؟ *

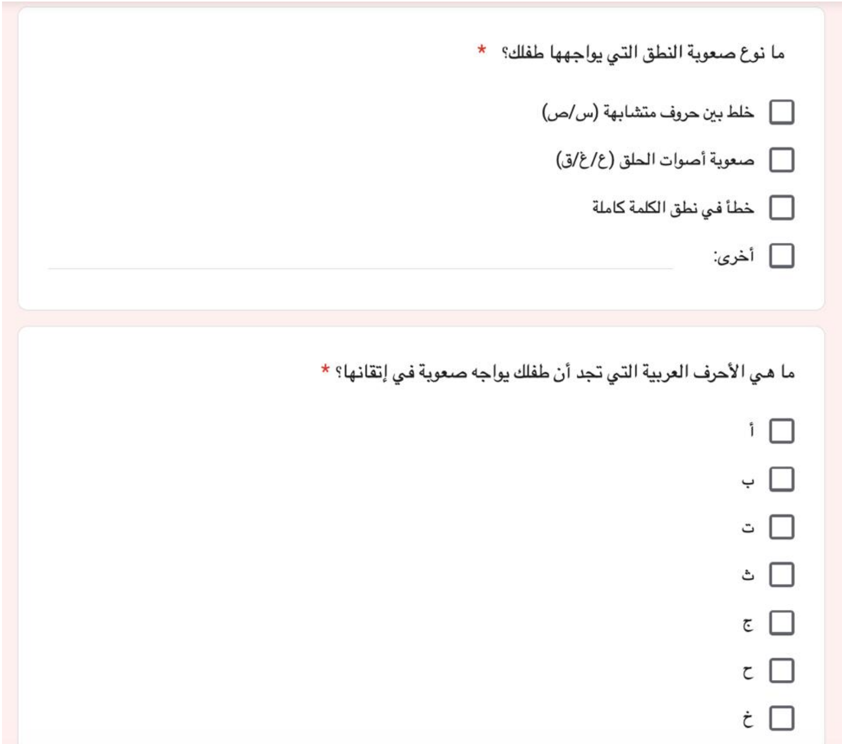
نعم

لا

إذا كانت الإجابة "نعم"، يرجى ذكر اسم التطبيق:

إجابتك

Figure 37: Questionnaire Q3, Q4, and Q5

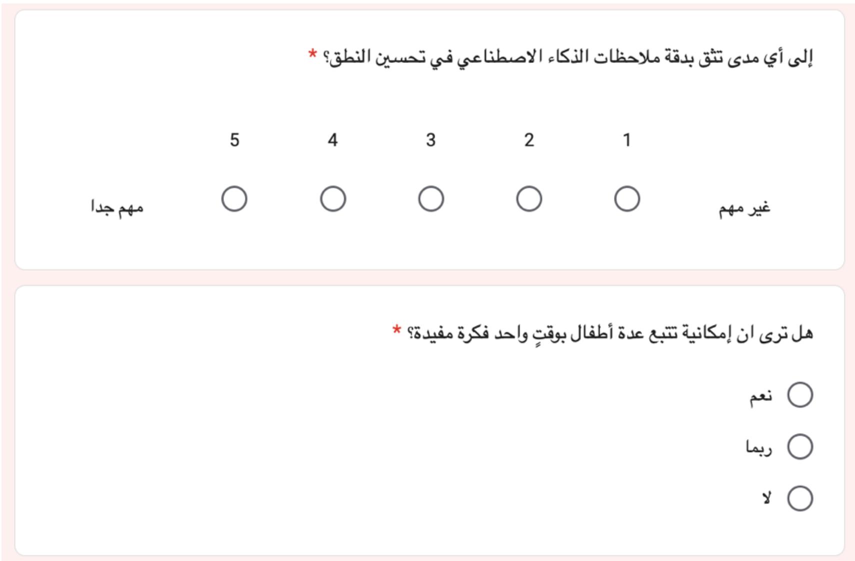

ما نوع صعوبة النطق التي يواجهها طفلك؟ *

- [ ] خلط بين حروف متشابهة (س/ص)
- [ ] صعوبة أصوات الحلق (ع/غ/ق)
- [ ] خطأ في نطق الكلمة كاملة
- [ ] أخرى:

ما هي الأحرف العربية التي تجد أن طفلك يواجه صعوبة في إتقانها؟ *

- [ ] أ
- [ ] ب
- [ ] ت
- [ ] ث
- [ ] ج
- [ ] ح
- [ ] خ

Figure 38: Questionnaire Q6 and Q7


إلى أي مدى تثق بدقة ملاحظات الذكاء الاصطناعي في تحسين النطق؟ *

	1	2	3	4	5	
غير مهم	○	○	○	○	○	مهم جدا

هل ترى ان إمكانية تتبع عدة أطفال بوقتٍ واحد فكرة مفيدة؟ *

- ○ نعم
- ○ ربما
- ○ لا

Figure 39: Questionnaire Q8 and Q9

ما مدى أهمية أن يتلقى طفلك تغذية راجعة فورية بعد نطق الكلمة ؟ *

غير مهم 1 2 3 4 5 مهم جدًا

هل تعتقد أن نظام النقاط التحفيزية يشجع الطفل على الاستمرار؟ *

- نعم
- ربما
- لا

Figure 40: Questionnaire Q10 and Q11

ما مدى أهمية تتبع تقدم الطفل داخل التطبيق؟ *

غير مهم 1 2 3 4 5 مهم جدًا

إذا كان هناك تطبيق يستخدم الذكاء الاصطناعي لمساعدة طفلك على تقييم و تحسين نطقه للغة العربية من خلال تقييمه وتقديم تغذية راجعة واضحة، فهل تفكر في استخدامه؟ *

- نعم
- ربما
- لا

Figure 41: Questionnaire Q12 and Q13

ما مدى أهمية العوامل التالية في التطبيق؟ *

	مهم جدا	مهم	محايد	غير مهم	غير مهم إطلاقًا
سهولة الاستخدام	○	○	○	○	○
واجهة مناسبة لعمر الطفل	○	○	○	○	○
وضوح التقارير المقدمة لولي الأمر	○	○	○	○	○
دقة تحليل النطق	○	○	○	○	○

Figure 42: Questionnaire Q14

ما الميزات التي تتوقعها من التطبيق؟ *

- ☐ تصحيح فوري للنطق
- ☐ شرح سبب الخطأ بالنطق
- ☐ تقارير أسبوعية
- ☐ خطة تعلم مخصصة
- ☐ نظام مكافآت
- ☐ تتبع تقدم الطفل
- ☐ دروس تقوية للغة العربية
- ☐ أخرى: ______

ما اقتراحاتك أو أفكارك لتحسين تطبيق يساعد الأطفال على تطوير نطقهم باللغة العربية؟

إجابتك

Figure 43: Questionnaire Q15 and Q16

8.1.1 Questionnaire Results

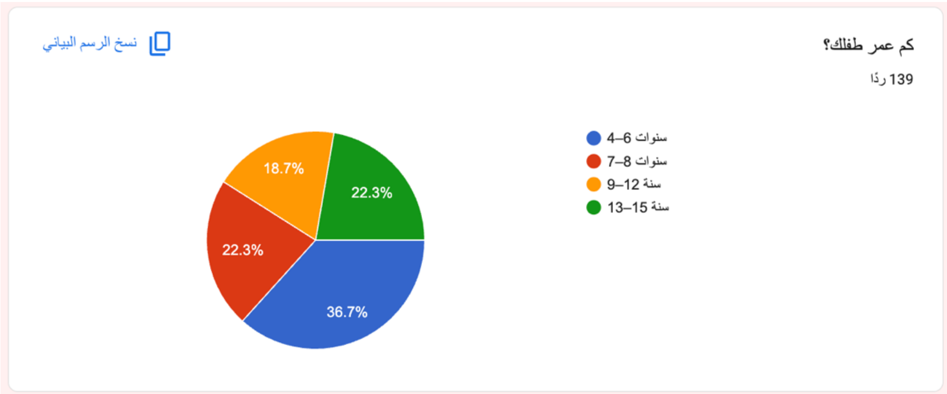


Figure 44: Q1 Answers Graph

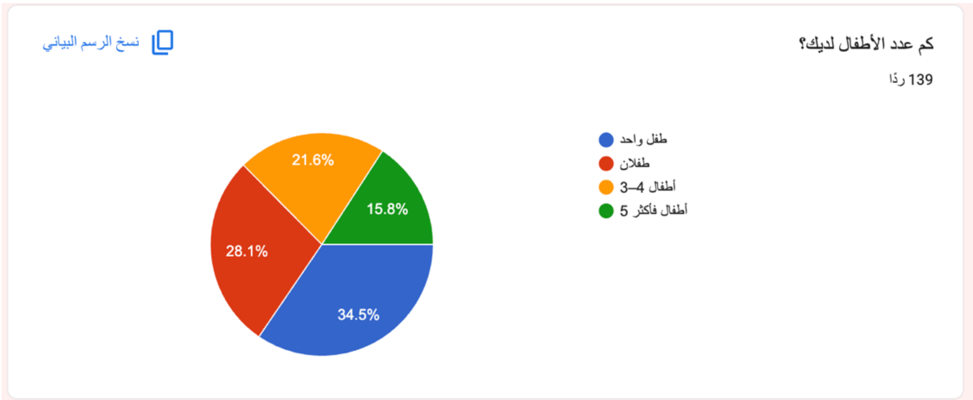


Figure 45: Q2 Answers Graph

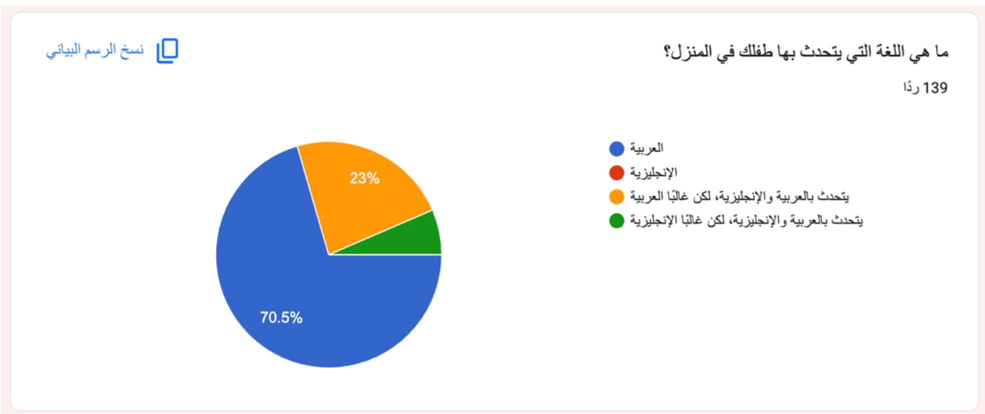


Figure 46: Q3 Answers Graph

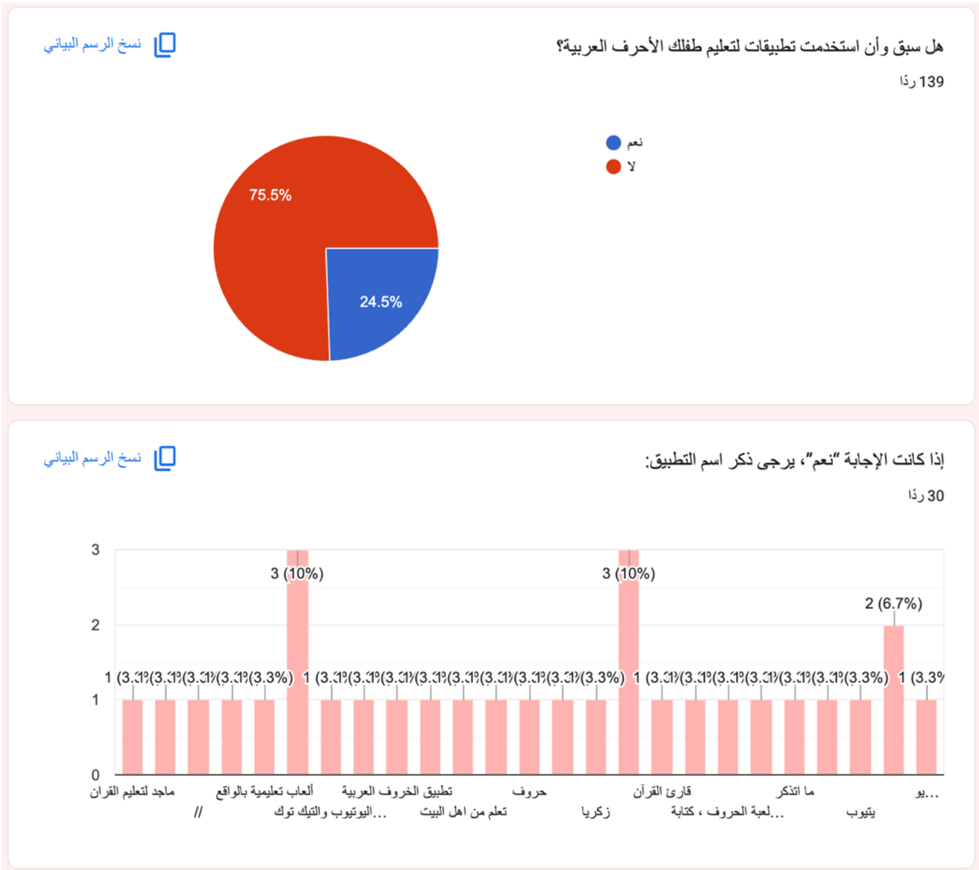


Figure 47: Q4 and Q5 Answers Graphs

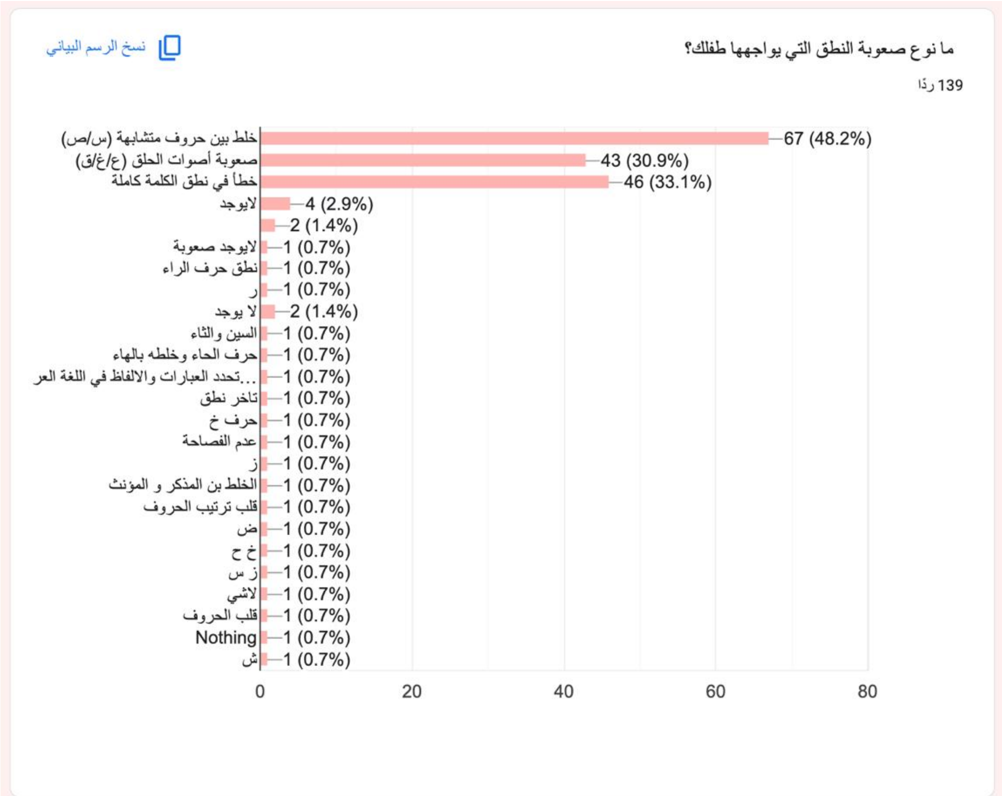


Figure 48: Q6 Answers Graph

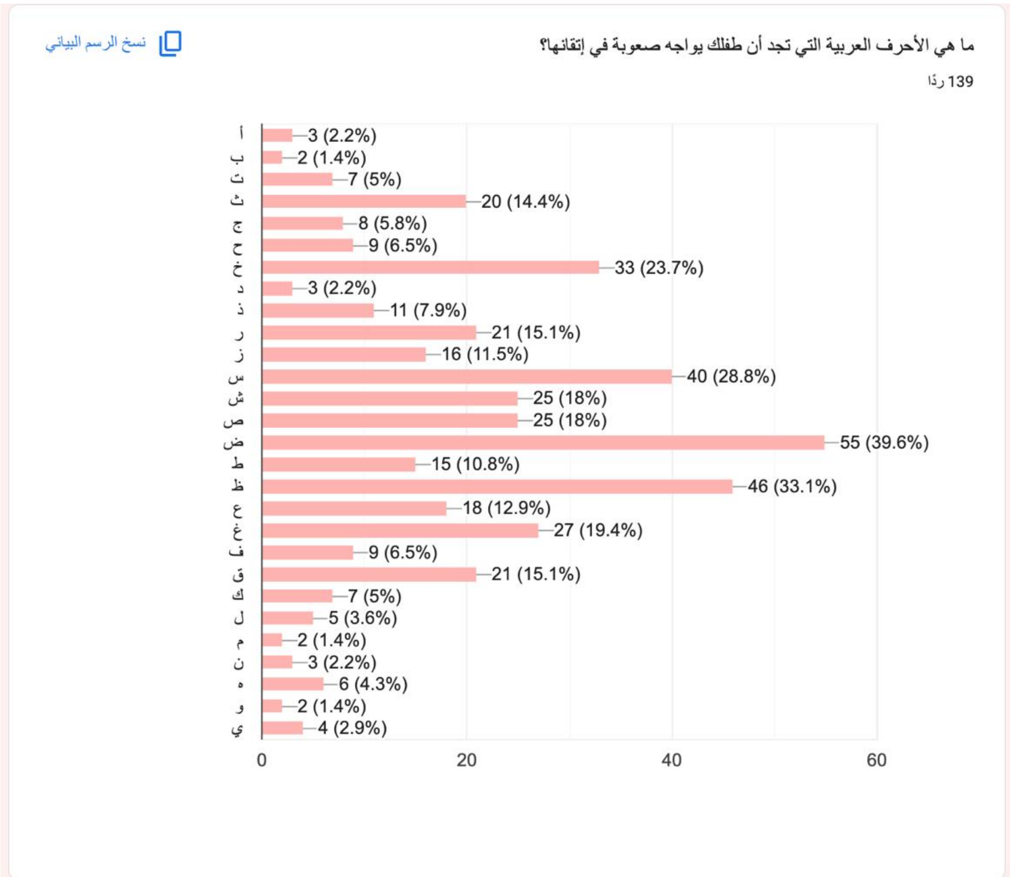


Figure 49: Q7 Answers Graph

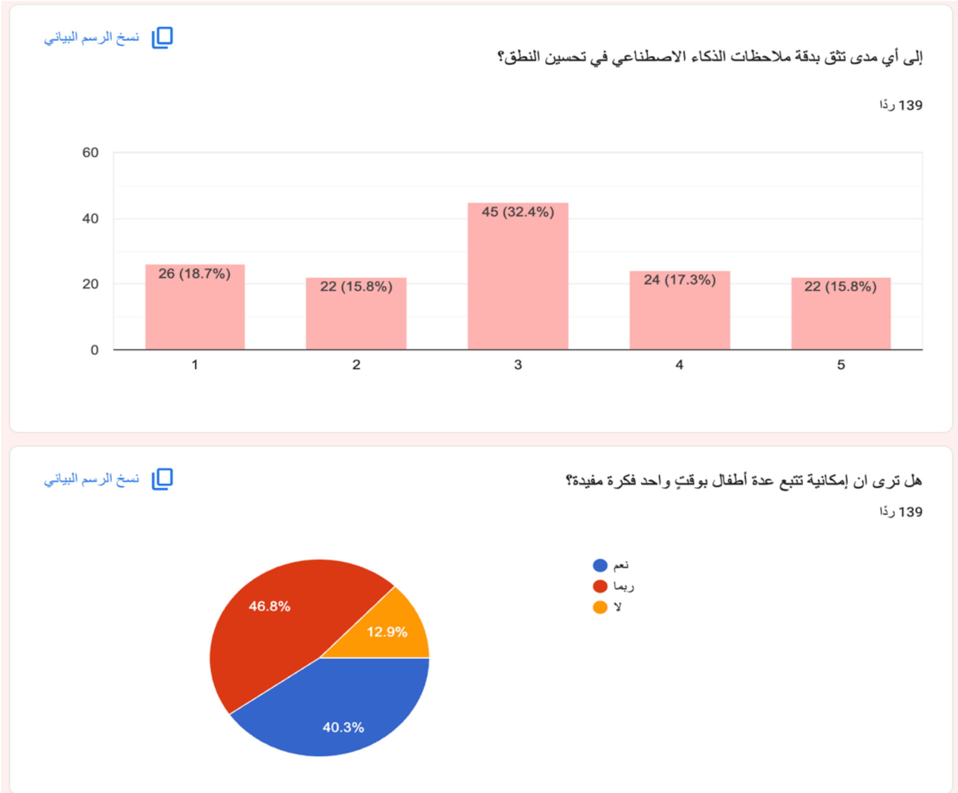


Figure 50: Q8 and Q9 Answers Graphs

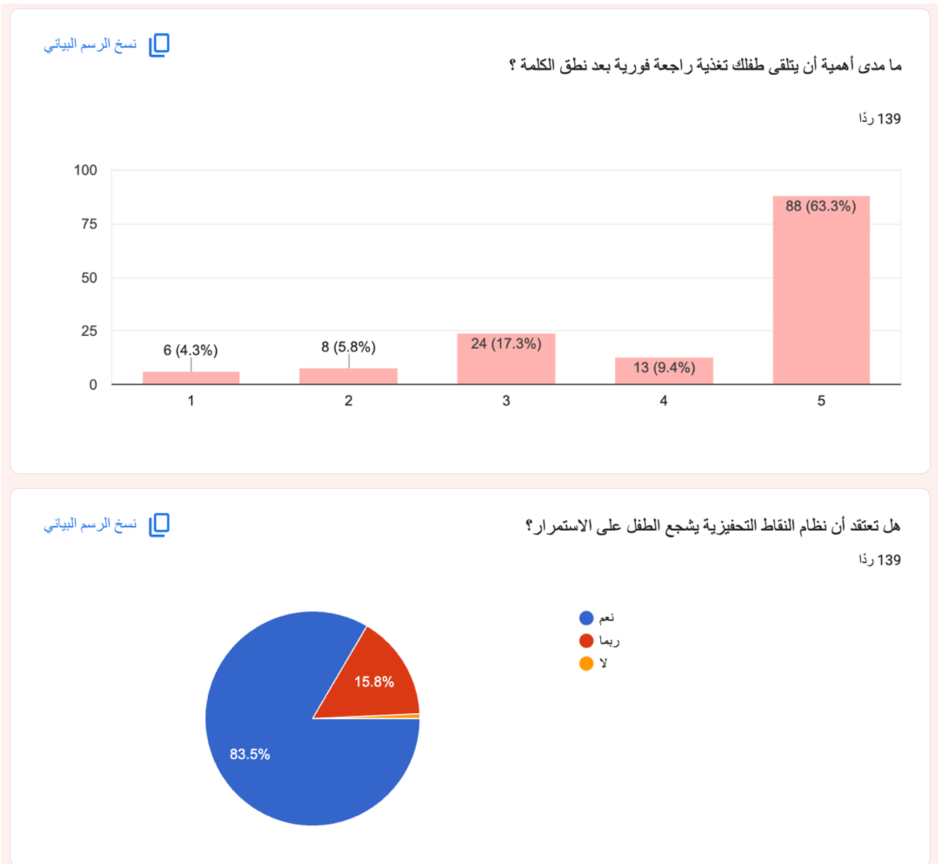


Figure 51: Q10 and Q11 Answers Graphs

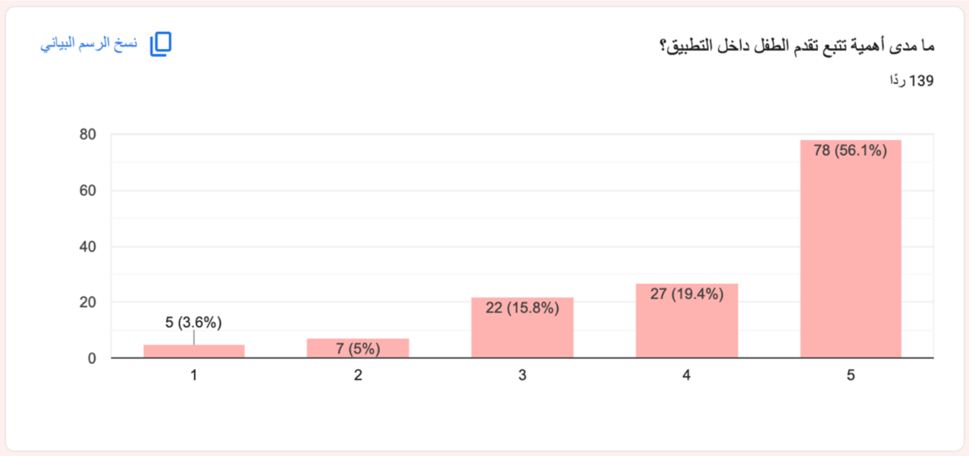


Figure 52: Q12 Answers Graph

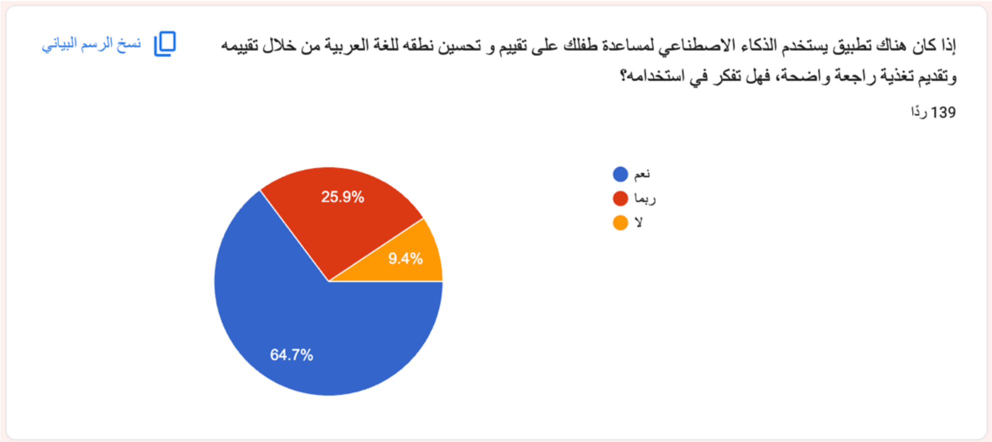


Figure 53:Q13 Answers Graph

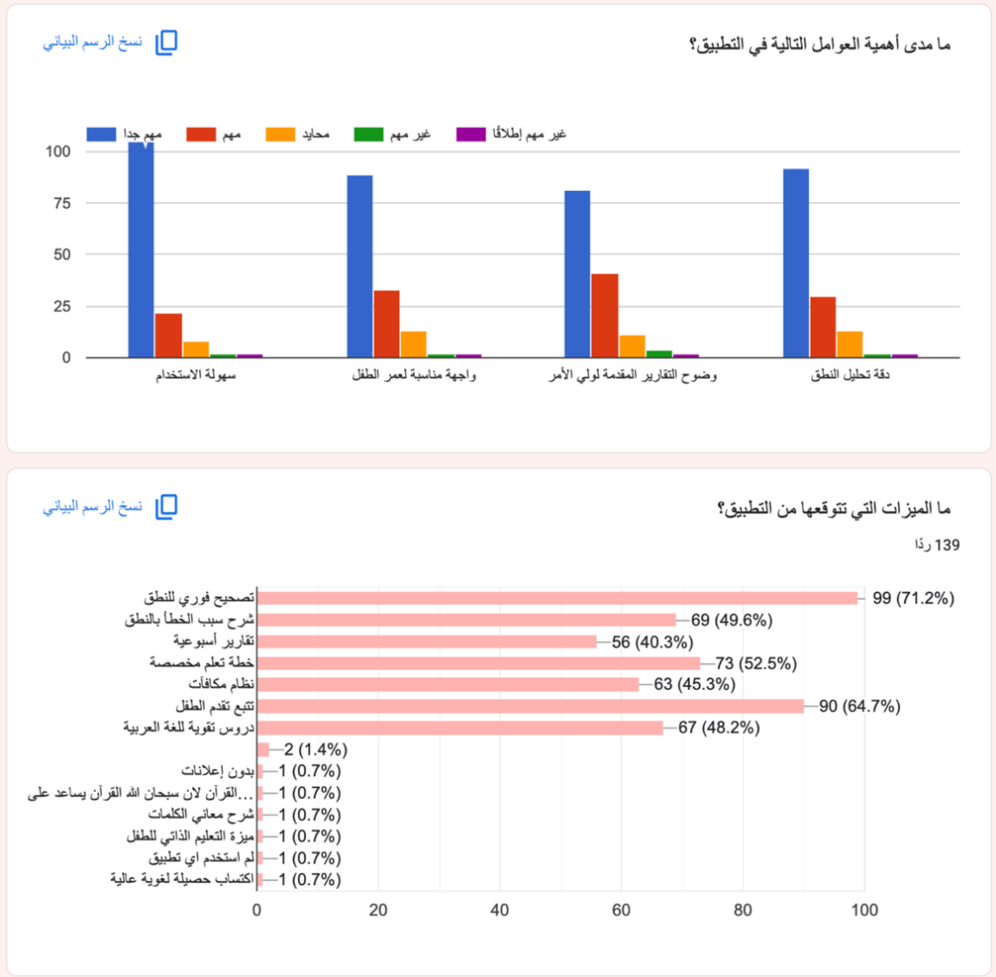


Figure 54:Q14 and Q15 Answers Graph

Figure 55: Q16 Answers Graph

8.2 Appendix B: Observation Results

Observation for child 1		
Child's Name:	Sahab	
Age:	7	
Date of observation:	16/2/2026	
Location:	At Home	
Duration of Observation:	25 min	
Observer's Name:	Joud Alzahrani	
Category	Questions	Answer
Ease of use	Can the child independently start the pronunciation exercise?	Yes. Sahab was able to independently start the pronunciation exercise without external assistance.
	Can the child record their voice without prior explanation?	Yes. She successfully recorded her voice by pressing the microphone icon without needing prior instruction
	Can the child navigate between screens easily?	Yes. She moved between screens smoothly and appeared comfortable using the interface.
Audio Recording Interaction	Does the child understand when to start and stop the recording?	Yes. She showed that she understood the recording process correctly. She pressed the microphone icon, started pronouncing the word at the right time, and waited until the app displayed her results after finishing.
	Does the child hesitate before recording?	No significant hesitation was observed. She appeared

		confident and initiated recording promptly.
Interaction with Feedback	Does the child pay attention to the corrective feedback provided?	Yes. She paid attention to the corrective feedback displayed by the application; however, she did not show interest in repeating the test.
	Is the child able to understand their performance based on the displayed feedback?	Yes. The child was able to clearly understand her performance based on the pronunciation feedback provided by the application.
Engagement level	Does the child show enthusiasm, or do they lose interest quickly?	She showed noticeable enthusiasm and remained engaged throughout the session.
	Does the child inquisitively ask questions during the activity?	She did not ask questions but remained focused and attentive while completing tasks.
	Does the child respond positively to the streak/reward system?	Yes. The child responded positively to the streak and reward system. She showed satisfaction when receiving points and achievements, and this appeared to motivate her to continue completing the activities.
Technical Issues	Did the child face difficulties understanding the system's messages?	No major difficulties were observed. She was able to understand most of the in-app instructions; however, minimal

		guidance was provided to help clarify certain messages due to her age.
	Was there any noticeable delays during the assessment process?	No. The application functioned smoothly without noticeable delays.

Table 9: Observation Results Child 1

Observation for child 2		
Child's Name:	Fahad	
Age:	9	
Date of observation:	15/2/2026	
Location:	Riyadh, Home environment	
Duration of Observation:	10 minutes	
Observer's Name:	Tala Alsheail	
Category	Questions	Answer
Ease of use	Can the child independently start the pronunciation exercise?	Partially. The child did not immediately identify how to start the exercise and spent some time exploring the interface before selecting the correct option.
	Can the child record their voice without prior explanation?	Yes, the child understood that pressing the microphone button would begin the recording without requiring prior explanation.
	Can the child navigate between screens easily?	Mostly yes. The child was able to move between screens, although he occasionally

		paused to locate the correct button.
Audio Recording Interaction	Does the child understand when to start and stop the recording?	Yes, the child understood that recording begins when the microphone is activated and ends once the word is completed.
	Does the child hesitate before recording?	Yes, the child showed slight hesitation before recording, likely wanting to ensure correct pronunciation.
Interaction with Feedback	Does the child pay attention to the corrective feedback provided?	Yes, the child paid attention to the corrective feedback and reviewed the results after each attempt.
	Is the child able to understand their performance based on the displayed feedback?	Partially. The child understood the result but needed clearer indication of which specific letter was mispronounced.
Engagement level	Does the child show enthusiasm, or do they lose interest quickly?	The child showed initial enthusiasm; however, attention slightly decreased when tasks became repetitive.
	Does the child inquisitively ask questions during the activity?	Yes. The child asked clarification questions such as which part of the word was incorrect.
	Does the child respond positively to the streak/reward system?	Yes. The reward system appeared to increase motivation and willingness to retry.

Technical Issues	Did the child face difficulties understanding the system's messages?	Partially. The child understood basic result messages such as "correct" or "incorrect," but appeared confused when system messages were longer or lacked clear visual cues.
	Was there any noticeable delays during the assessment process?	No significant delays were observed; however, the child appeared slightly confused during brief waiting moments before the results were displayed.

Table 10: Observation Results Child 2

Observation for child 3		
Child's Name:	Mohammed	
Age:	10	
Date of observation:	17/2/2026	
Location:	Riyadh, Home Environment	
Duration of Observation:	6 Minutes	
Observer's Name:	Norah Alkathiri	
Category	Questions	Answer
Ease of use	Can the child independently start the pronunciation exercise?	Yes, the pronunciation exercise seemed to have started automatically so the child didn't have to search for it in the app.
	Can the child record their voice without prior explanation?	Yes, the child did not need any prior guidance to understand how to record their audio.

	Can the child navigate between screens easily?	Yes, the child was able to navigate through the app smoothly.
Audio Recording Interaction	Does the child understand when to start and stop the recording?	Partially, the child understood how to start the recording but seemed to be facing some difficulties with stopping it.
	Does the child hesitate before recording?	No, the child didn't seem to be nervous or hesitant during the exam in general.
Interaction with Feedback	Does the child pay attention to the corrective feedback provided?	Partially, some of the feedback was unclear to the child as it only showed the correct answer but didn't compare between the child's answer and the correct one.
	Is the child able to understand their performance based on the displayed feedback?	Somewhat, after clarifying the child's exact mistakes, they seemed to have understood how they were performing.
Engagement level	Does the child show enthusiasm, or do they lose interest quickly?	Yes, the child shows excitement throughout the process of the examination.
	Does the child inquisitively ask questions during the activity?	Yes, but only when they received their feedback since they did not understand it entirely.
	Does the child respond positively to the streak/reward system?	The child showed enthusiasm initially but seemed to lose it at the end of the examination.

Technical Issues	Did the child face difficulties understanding the system's messages?	Partially, the child understood most of the system's messages except for some question feedback that was not clear.
	Were there any noticeable delays during the assessment process?	No, the child did not experience any delays during the examination process or while navigating between the different interfaces in the app.

Table 11: Observation Results Child 3

Observation for child 4		
Child's Name:	Alma	
Age:	12	
Date of observation:	16/2/2026	
Location:	Riyadh, in her grandmother's house	
Duration of Observation:	5 min	
Observer's Name:	Walah Alsaeed	
Category	Questions	Answer
Ease of use	Can the child independently start the pronunciation exercise?	Yes, the child was able to independently initiate the pronunciation exercise without any external assistance by pressing the microphone button and beginning to speak.
	Can the child record their voice without prior explanation?	Yes, no prior explanation was required, as the child read the instructions and pressed the recording button to complete the pronunciation test.

	Can the child navigate between screens easily?	Yes, the navigation was clear to the child, as she was able to move from the test page to the results page and return to the exercises page without any assistance or questions.
Audio Recording Interaction	Does the child understand when to start and stop the recording?	Yes, the child clearly understood when to start and stop the recording. After pressing the recording button, she began pronouncing the word, and upon completion, the results were displayed immediately.
	Does the child hesitate before recording?	No, the child confidently initiated the recording without any hesitation.
Interaction with Feedback	Does the child pay attention to the corrective feedback provided?	Yes, the child paid attention to the feedback and, when a word was mispronounced, immediately repeated the pronunciation test in order to pronounce it correctly.
	Is the child able to understand their performance based on the displayed feedback?	Yes, the child was able to easily understand her performance based on the pronunciation feedback. However, regarding the final overall feedback for the entire test, which included

		writing and listening skills as part of the score, she did not pay attention to it and was therefore unable to understand how she performed.
Engagement level	Does the child show enthusiasm, or do they lose interest quickly?	Yes, the child showed enthusiasm in completing the test and maintained a high level of attention throughout.
	Does the child inquisitively ask questions during the activity?	No, the child remained quiet throughout the entire test, completing each task sequentially without asking any questions.
	Does the child respond positively to the streak/reward system?	No, the child did not show interest in the streak or reward system and did not pay attention to the points earned.
Technical Issues	Did the child face difficulties understanding the system's messages?	The child did not fully understand the notifications presented in the initial tutorial of the application, as she skipped through them; however, all other notifications that appeared during the test were understood.
	Was there any noticeable delays during the assessment process?	No, there were no noticeable delays during the assessment process. After the account was created by the observer, the

		child took the device and began the test smoothly.

Table 12: Observation Results Child 4

Observation for child 5		
Child's Name:	Abdullah	
Age:	13	
Date of observation:	18/2/2026	
Location:	Riyadh, Home Environment	
Duration of Observation:	9 Minutes	
Observer's Name:	Norah Alkathiri	
Category	Questions	Answer
Ease of use	Can the child independently start the pronunciation exercise?	Not entirely, the child needed some time to figure out the exact way to get into the exercise first.
	Can the child record their voice without prior explanation?	Yes, once the child knew how to start the exercise, they faced no further troubles.
	Can the child navigate between screens easily?	Yes, the navigation between screen/interfaces was smooth since it was all clarified in the bottom icon bar in the app.
Audio Recording Interaction	Does the child understand when to start and stop the recording?	Yes, the pronunciation recording process was easy to understand for the child.
	Does the child hesitate before recording?	No, the child showed no signs of hesitation once they recognized the word they were asked to pronounce.

Interaction with Feedback	Does the child pay attention to the corrective feedback provided?	Yes, the child has also used the feedback to avoid mistakes in further exercises.
	Is the child able to understand their performance based on the displayed feedback?	Yes, since the app clarified if the answer was wrong or right by a visible message along with a sound notification.
Engagement level	Does the child show enthusiasm, or do they lose interest quickly?	The child began to lose interest by the end of the exercise.
	Does the child inquisitively ask questions during the activity?	Yes, the child was confused with some of the questions and what they required.
	Does the child respond positively to the streak/reward system?	Yes, the child seemed to have cheer up once they've gotten the question correctly.
Technical Issues	Did the child face difficulties understanding the system's messages?	Yes, as mentioned previously some of the questions were unclear. For example, a question said "اختر العنوان" when there was no title that could've shown the correct answer.
	Were there any noticeable delays during the assessment process?	Yes, there were some delays during the transition between different questions, and the speaker icon which plays the audio containing a word associated with the question, seemed to have some delay before playing the sound.

Table 13: Observation Results Child 5

Observation for child 6		
Child's Name:	Atheer	
Age:	6	
Date of observation:	19/2/2026	
Location:	Riyadh, Home Environment	
Duration of Observation:	11 Minutes	
Observer's Name:	Tala Alsheail	
Category	Questions	Answer
Ease of use	Can the child independently start the pronunciation exercise?	No. The child required guidance to identify the correct button to start the exercise and did not initiate it independently.
	Can the child record their voice without prior explanation?	No. The child appeared unsure about how to begin recording and needed instructions on using the microphone button.
	Can the child navigate between screens easily?	Partially. The child was able to move between screens with some assistance but seemed confused by multiple options on the interface.
Audio Recording Interaction	Does the child understand when to start and stop the recording?	Not immediately. The child showed confusion during the first attempt and required demonstration to understand the recording process.
	Does the child hesitate before recording?	Yes. The child hesitated and appeared shy or uncertain before recording her voice.

Interaction with Feedback	Does the child pay attention to the corrective feedback provided?	Partially. The child noticed whether the result was correct or incorrect but did not fully engage with detailed feedback.
	Is the child able to understand their performance based on the displayed feedback?	No. The child struggled to understand written feedback and relied mainly on visual indicators such as colors or symbols.
Engagement level	Does the child show enthusiasm, or do they lose interest quickly?	The child showed initial excitement but lost focus more quickly.
	Does the child inquisitively ask questions during the activity?	No. The child did not ask questions and instead looked to the observer for reassurance or confirmation.
	Does the child respond positively to the streak/reward system?	Yes. The child responded positively to visual rewards such as stars and animations.
Technical Issues	Did the child face difficulties understanding the system's messages?	Yes. The child had difficulty understanding text-based messages and benefited more from visual cues.
	Was there any noticeable delays during the assessment process?	No significant delays were observed; however, the child became impatient during brief waiting moments.

Table 14: Observation Results Child 6

Observation for child 7	
Child's Name:	Aleen

Age:	11	
Date of observation:	20/2/2026	
Location:	Riyadh, in her grandmother's house	
Duration of Observation:	6 min	
Observer's Name:	Walah Alsaeed	
Category	**Questions**	**Answer**
Ease of use	Can the child independently start the pronunciation exercise?	Yes, the child was able to independently initiate the pronunciation exercise without any external assistance.
	Can the child record their voice without prior explanation?	Yes, no prior explanation was required, the child pressed the recording button and completed the pronunciation test.
	Can the child navigate between screens easily?	The child initially experienced some difficulty in identifying where to begin the first exercise and required minimal guidance; however, she was later able to navigate between the screens independently.
Audio Recording Interaction	Does the child understand when to start and stop the recording?	The child clearly understood when to start and stop the recording. She began saying the word after pressing the recording button, and the results appeared immediately after she finished.

	Does the child hesitate before recording?	No, the child confidently initiated the recording without any hesitation.
Interaction with Feedback	Does the child pay attention to the corrective feedback provided?	Yes, the child paid attention to the corrective feedback and was aware when she mispronounced a word.
	Is the child able to understand their performance based on the displayed feedback?	Yes, the child understood her performance based on the feedback that appeared after each question. However, she did not pay attention to the final overall feedback until she was asked about the results page and what she understood from it.
Engagement level	Does the child show enthusiasm, or do they lose interest quickly?	Yes, the child showed enthusiasm in completing the test and maintained a high level of attention throughout.
	Does the child inquisitively ask questions during the activity?	No, the child did not ask questions during the activity, as she completed the exercises quietly.
	Does the child respond positively to the streak/reward system?	No, the child did not show interest in the streak or reward system and did not pay attention to the points earned.

Technical Issues	Did the child face difficulties understanding the system's messages?	The child did not carefully read the notifications shown in the initial tutorial, as she skipped through them. However, she was able to understand the notifications that appeared during the test.
	Was there any noticeable delays during the assessment process?	No, there were no noticeable delays during the assessment process. After the account was created by the observer, the child took the device and began the test smoothly.

Table 15: Observation Results Child 7

Observation for child 8		
Child's Name:	Adeem	
Age:	7	
Date of observation:	20/2/2026	
Location:	Riyadh, in her grandmother's house	
Duration of Observation:	6 min	
Observer's Name:	Walah Alsaeed	
Category	Questions	Answer
Ease of use	Can the child independently start the pronunciation exercise?	Yes, the child was able to independently initiate the pronunciation exercise without any external assistance, but she was a little bit hesitated before pressing the record button.
	Can the child record their voice without prior explanation?	Yes, no prior explanation was required, the child pressed the

		recording button and completed the pronunciation test.
	Can the child navigate between screens easily?	No, the child did not know where to begin, and the observer needed to start the exercise for her.
Audio Recording Interaction	Does the child understand when to start and stop the recording?	The child knew when to start the recording; however, she did not know when to stop it. She sometimes stopped the recording before finishing and clicked randomly, which caused her to skip before the results were displayed by mistake. The observer needed to provide instructions on when to stop and assisted her once by stopping the recording.
	Does the child hesitate before recording?	Yes, when the pronunciation exercise was shown for the first time, the child hesitated briefly before starting the recording; however, she began the recording independently after a few seconds.
Interaction with Feedback	Does the child pay attention to the corrective feedback provided?	Yes, the child paid attention to the corrective feedback and was aware when she mispronounced a word.

	Is the child able to understand their performance based on the displayed feedback?	Yes, the child was able to understand her performance from the feedback provided after each question. However, she did not focus on the final overall feedback unless she was specifically asked about the results page and her understanding of it.
Engagement level	Does the child show enthusiasm, or do they lose interest quickly?	Yes, the child showed enthusiasm in completing the test and maintained a high level of attention throughout.
	Does the child inquisitively ask questions during the activity?	Yes, the child asked a few questions when certain exercises were not clear to her, such as listening or more difficult writing exercises.
	Does the child respond positively to the streak/reward system?	No, the child did not show interest in the streak or reward system and did not pay attention to the points earned.
Technical Issues	Did the child face difficulties understanding the system's messages?	Yes, the child experienced some difficulty understanding certain system messages. For example, when she attempted to access an exercise that was not appropriate for her level, she did not understand why it was not opening.

	Was there any noticeable delays during the assessment process?	No, there were no noticeable delays during the assessment process.

Table 16: Observation Results Child 8

8.3 Appendix C: Tools and Environment

This section describes the tools and development environment used to manage and implement the Graduation Project. To ensure effective project management, documentation, collaboration, and version control, the team utilized Jira, Confluence, and GitHub. These tools enabled structured planning, organized documentation, transparent progress tracking, and efficient source code management throughout the project development.

Jira Software

We created a Jira project with the official name format:
2026-GP1-G2**.**

The following tasks were completed in Jira:

- Created the Product Backlog and populated it with all user stories covering both GP1 and GP2 phases.
- Organized and prioritized backlog items according to project requirements.
- Used Jira to track sprint progress and manage tasks systematically.

[Jira Link]

Confluence

We created a Confluence space and linked it directly to our Jira project to maintain organized and centralized documentation.

The Confluence space was structured according to the Jira Guide and included:

- Meeting Notes

- Sprint Reviews
- Submitted Documents

Confluence Link

GitHub

We created a GitHub repository under the name:
2026-GP1-G2**.**

GitHub is going to be used for:

- Managing and storing the project source code
- Version control
- Tracking code updates and collaboration among team members

GitHub Link